RWTH AACHEN UNIVERSITY

Computer Vision – Lecture 11

CNNs IV

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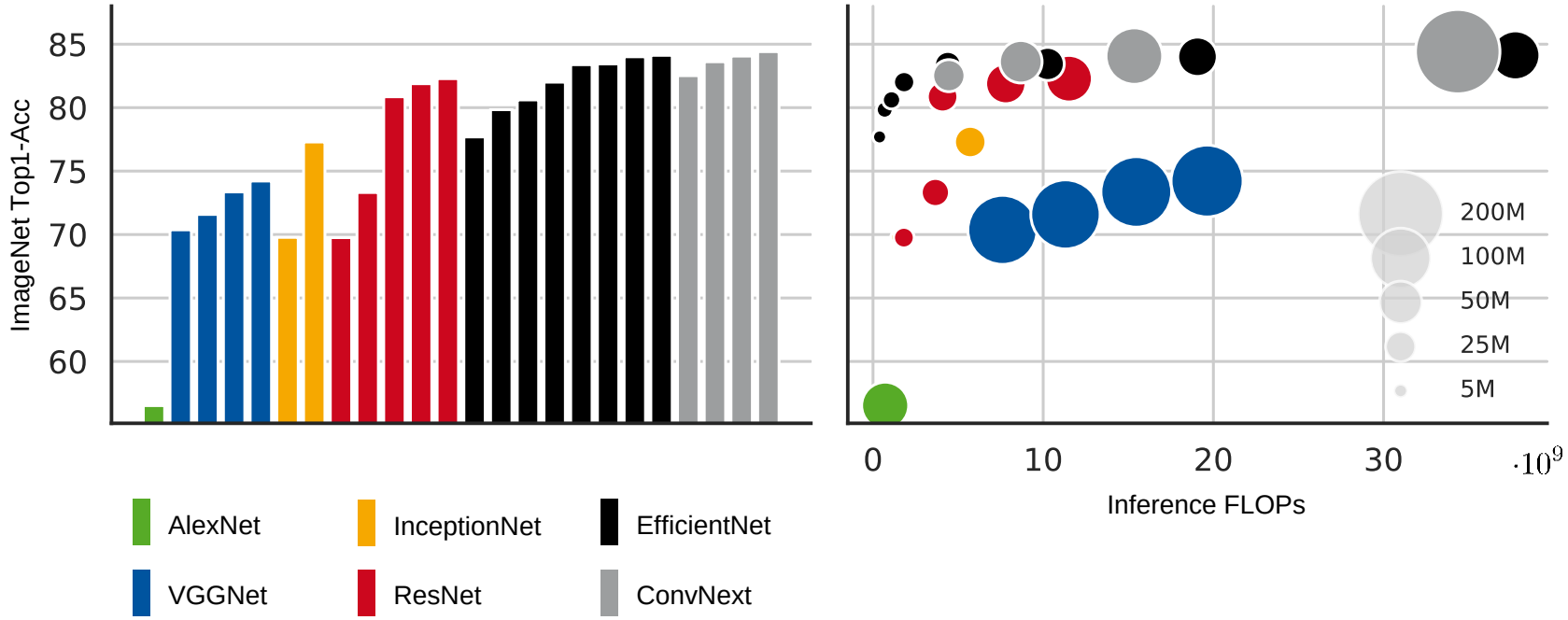
Course Outline

- Image Processing Basics
- Segmentation & Grouping
- Object Recognition & Categorization
  - Sliding Window based Object Detection
- Deep Learning for Vision
  - Convolutional Neural Networks (CNNs)
  - Deep Learning Background
  - CNNs for Object Detection
  - CNNs for Semantic Segmentation
  - CNNs for Matching
  - Transformers
- 3D Reconstruction

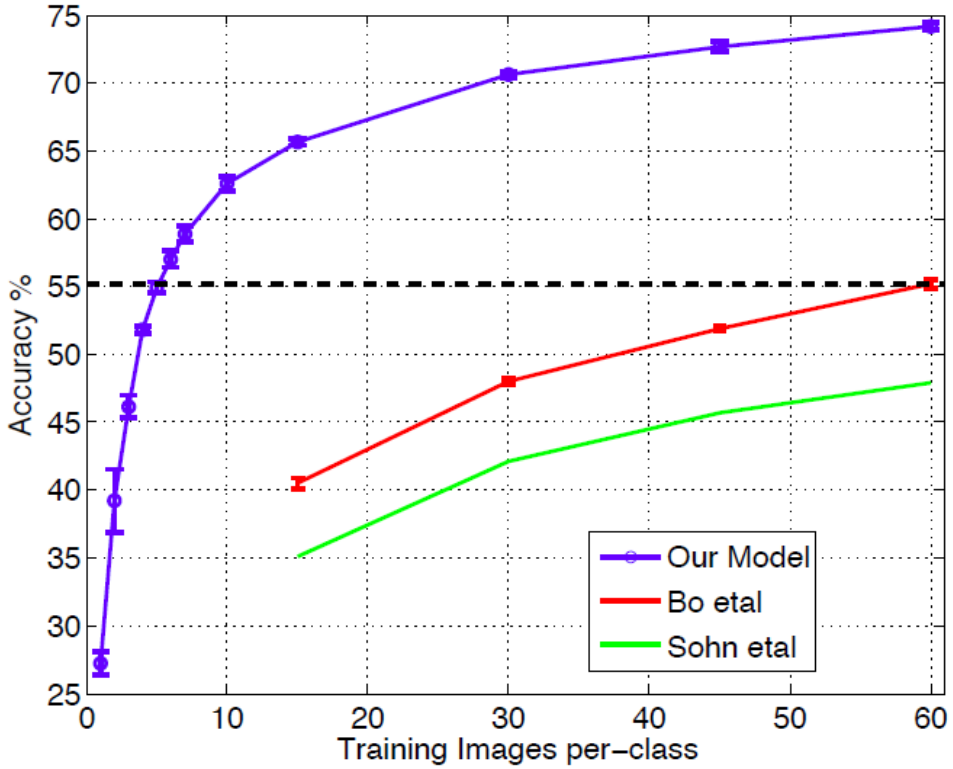
Recap: CNN Backbone Architectures



Models trained on ImageNet-1K, as available in torchvision v0.17.2

- g

The Learned Features are Generic



state of the art level (pre-CNN)

- Experiment: feature transfer
  - Train AlexNet-like network on ImageNet
  - Chop off last layer and train classification layer on CalTech256

  ⇒ State of the art accuracy already with only 6 training images!

Image source: M. Zeiler, R. Fergus

Transfer Learning with CNNs

1. Train on ImageNet

2. If small dataset: fix all weights (treat CNN as fixed feature extractor), retrain only the classifier

I.e., swap the Softmax layer at the end

Slide credit: Andrej Karpathy

Transfer Learning with CNNs

1. Train on ImageNet

3. If you have medium sized dataset, "finetune" instead: use the old weights as initialization, train the full network or only some of the higher layers.

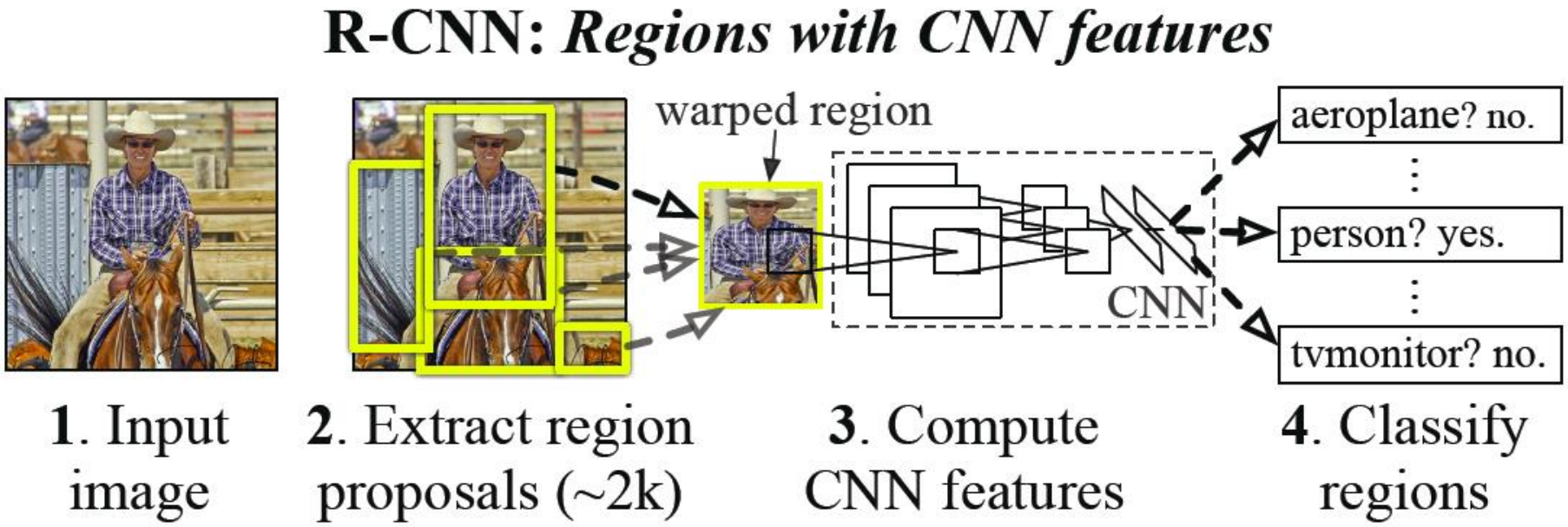
Retrain bigger portion of the network

Slide credit: Andrej Karpathy

Topics of This Lecture

- CNNs for Object Detection
  - The R-CNN Family
  - The YOLO Family
- CNNs for Segmentation
  - Fully Convolutional Networks (FCN)
  - Encoder-Decoder architecture
  - Transpose convolutions
  - Skip connections
- CNNs for Human Pose Estimation
  - 2D Body pose estimation
  - 3D Body pose estimation
- CNNs for Matching
  - Siamese networks
  - Triplet loss

Object Detection: R-CNN Idea

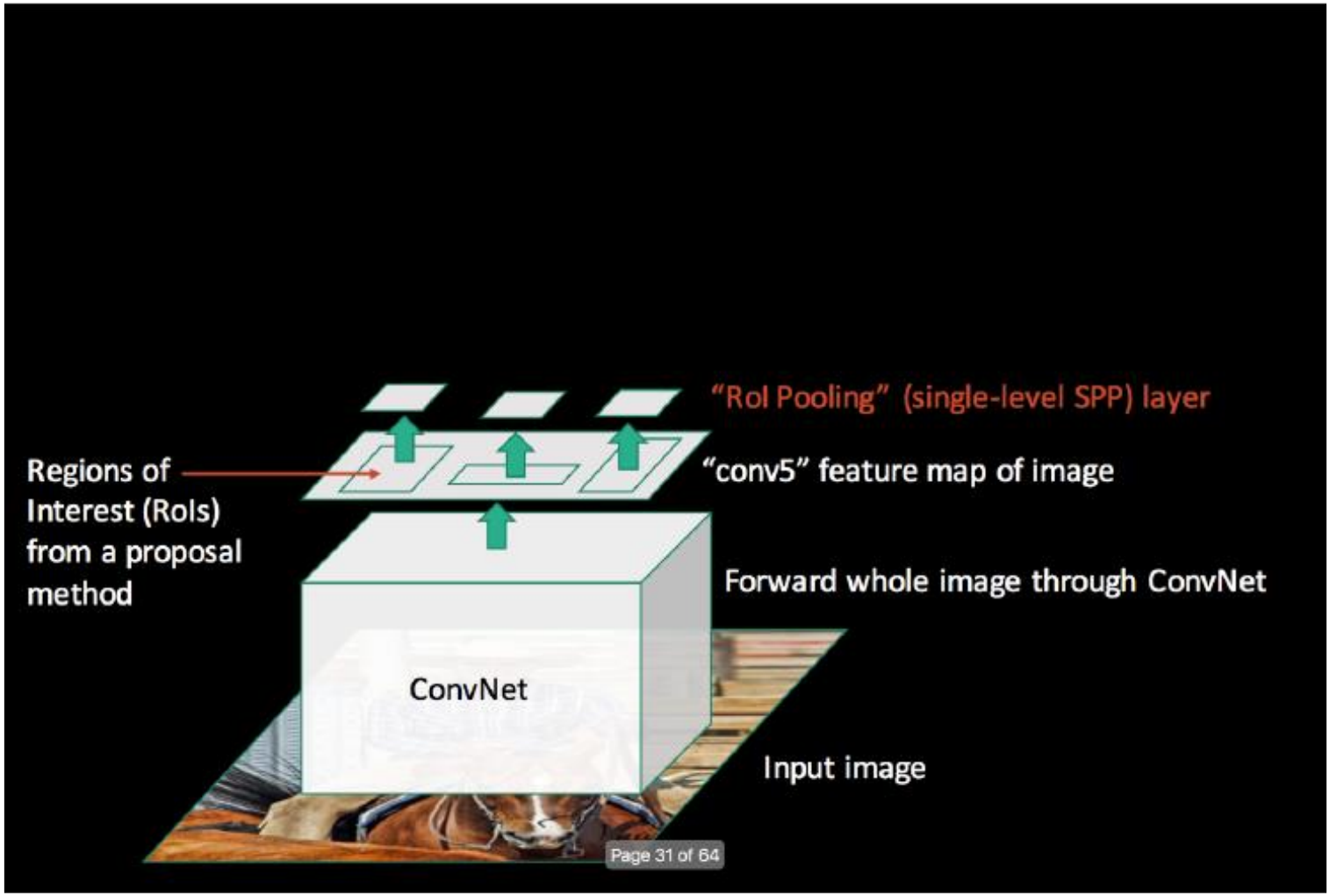


- Results on PASCAL VOC Detection benchmark
  - Pre-CNN state of the art: 35.1% mAP Selective Search
    33.4% mAP DPM
  - R-CNN: 53.7% mAP

R. Girshick, J. Donahue, T. Darrell, and J. Malik, Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation, CVPR 2014

Improved Version: Fast R-CNN

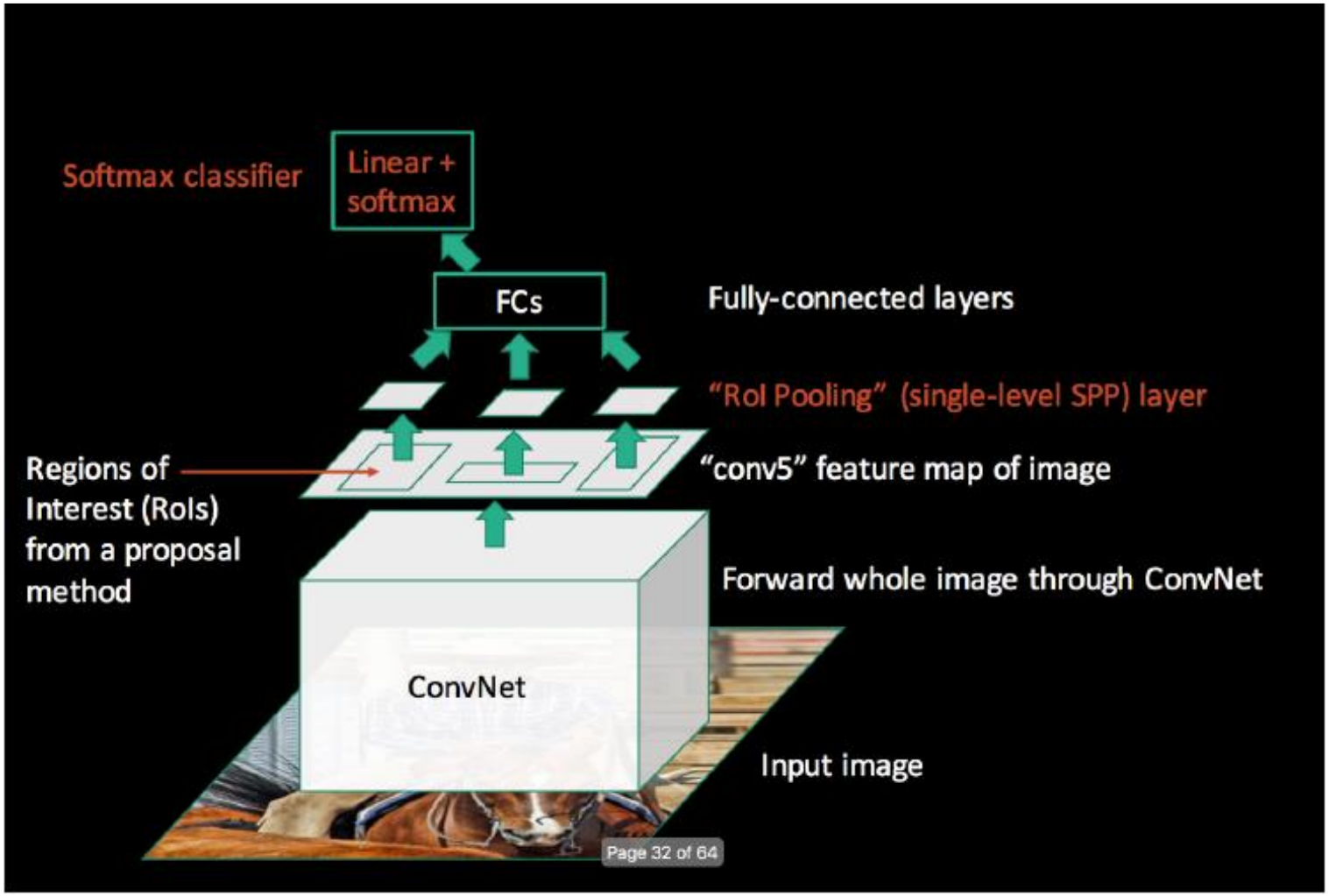
- Forward Pass



Slide credit: Ross Girshick

Fast R-CNN

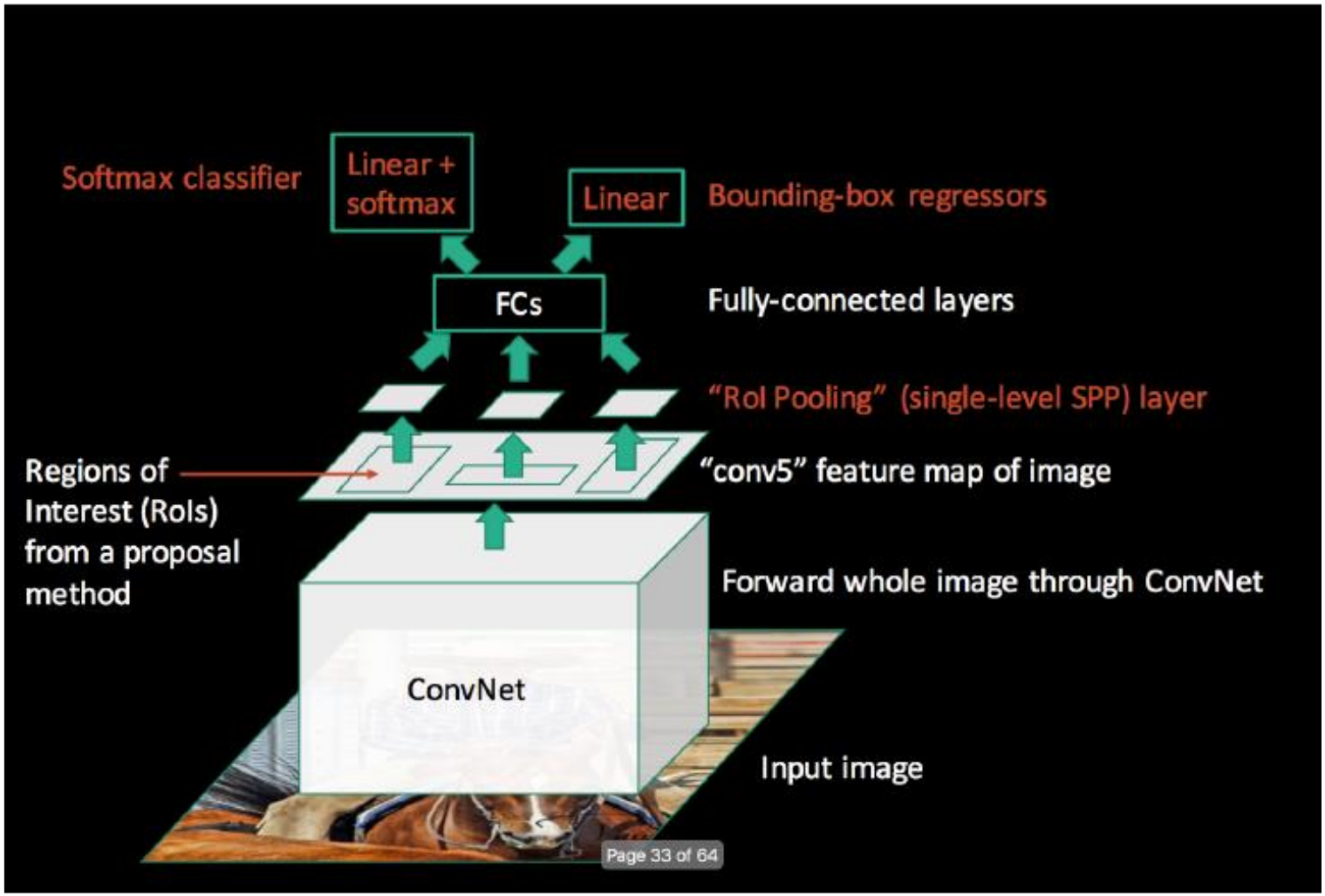
- Forward Pass



Slide credit: Ross Girshick

Fast R-CNN

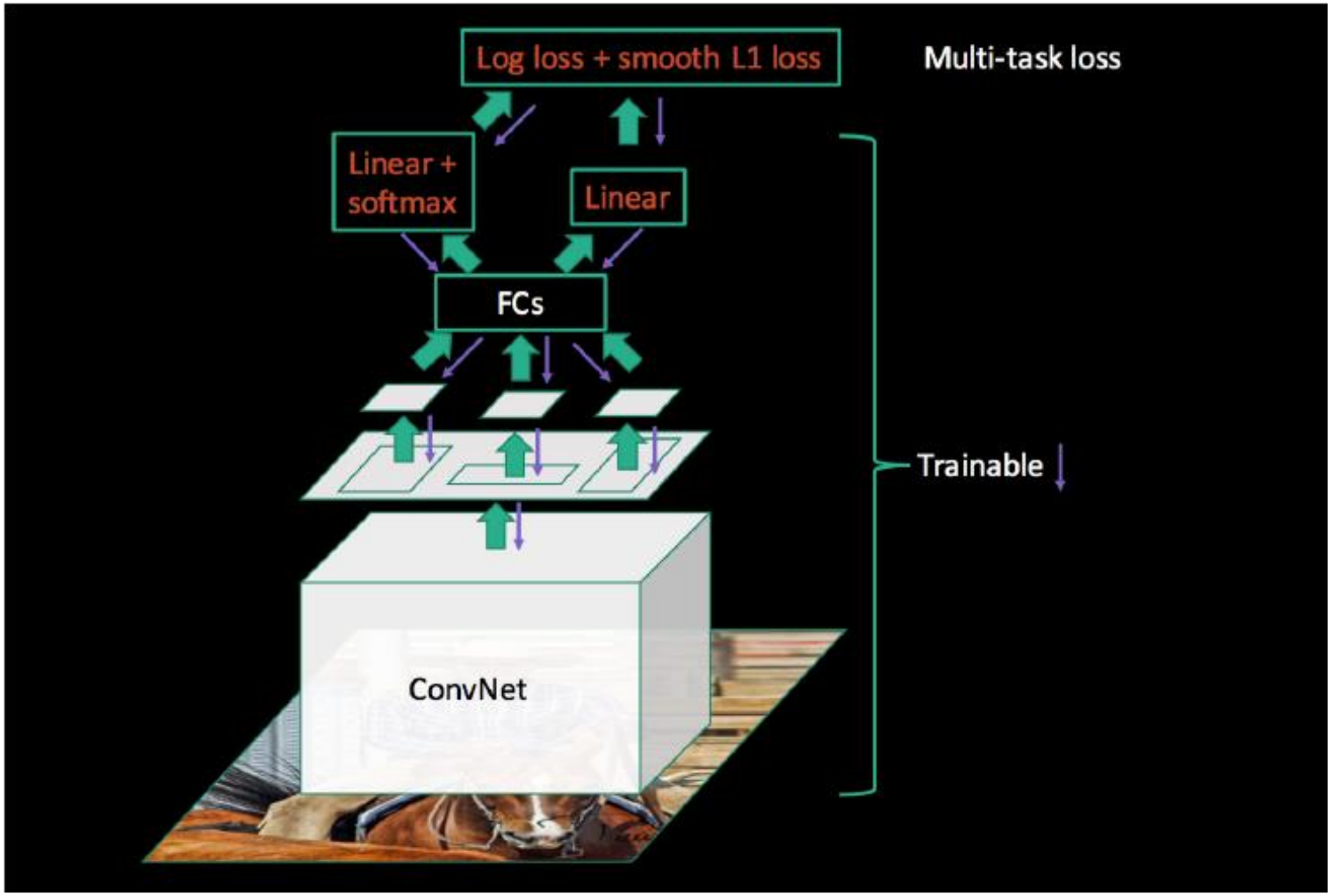
- Forward Pass



Slide credit: Ross Girshick

Fast R-CNN Training

- Backward Pass



Slide credit: Ross Girshick

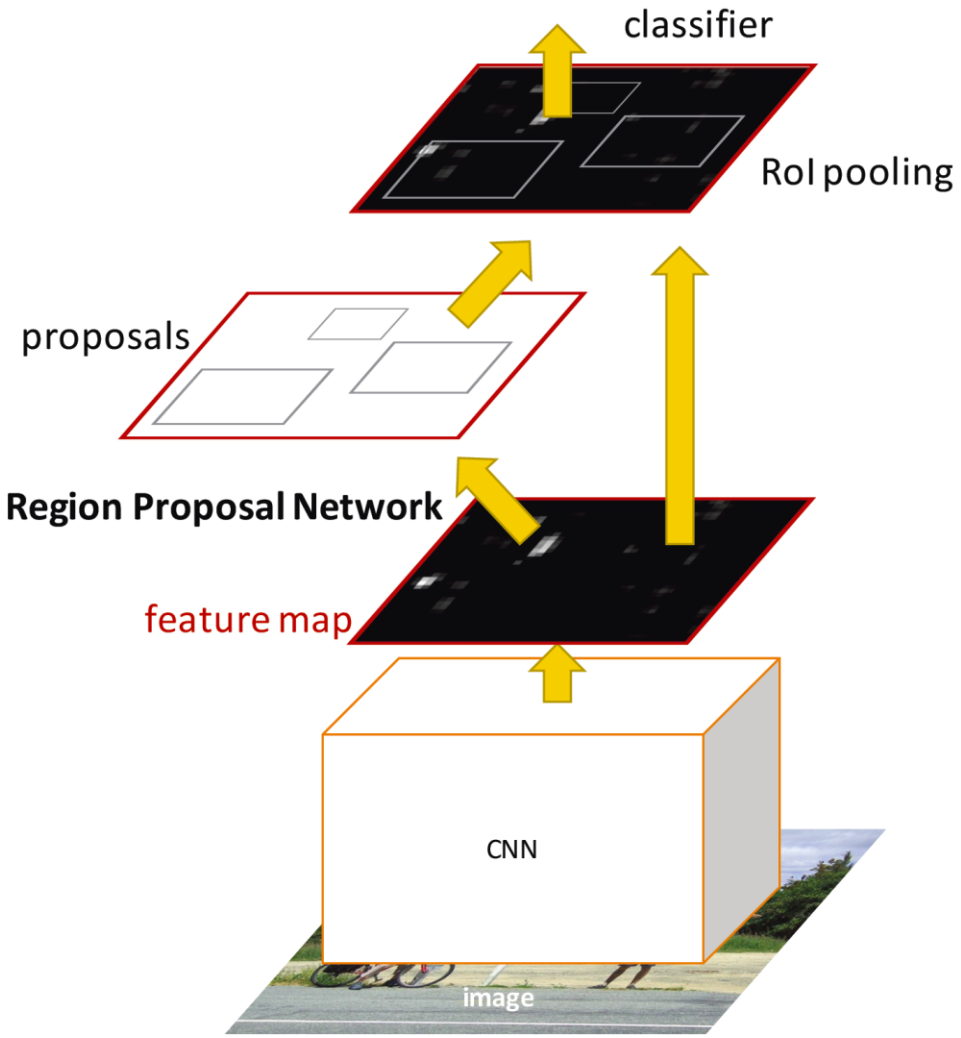
Region Proposal Networks (RPN)

- Idea
  - Remove dependence on external region proposal algorithm.
  - Instead, infer region proposals from same CNN.

  ⇒ Feature sharing

  ⇒ Object detection in a single pass becomes possible.

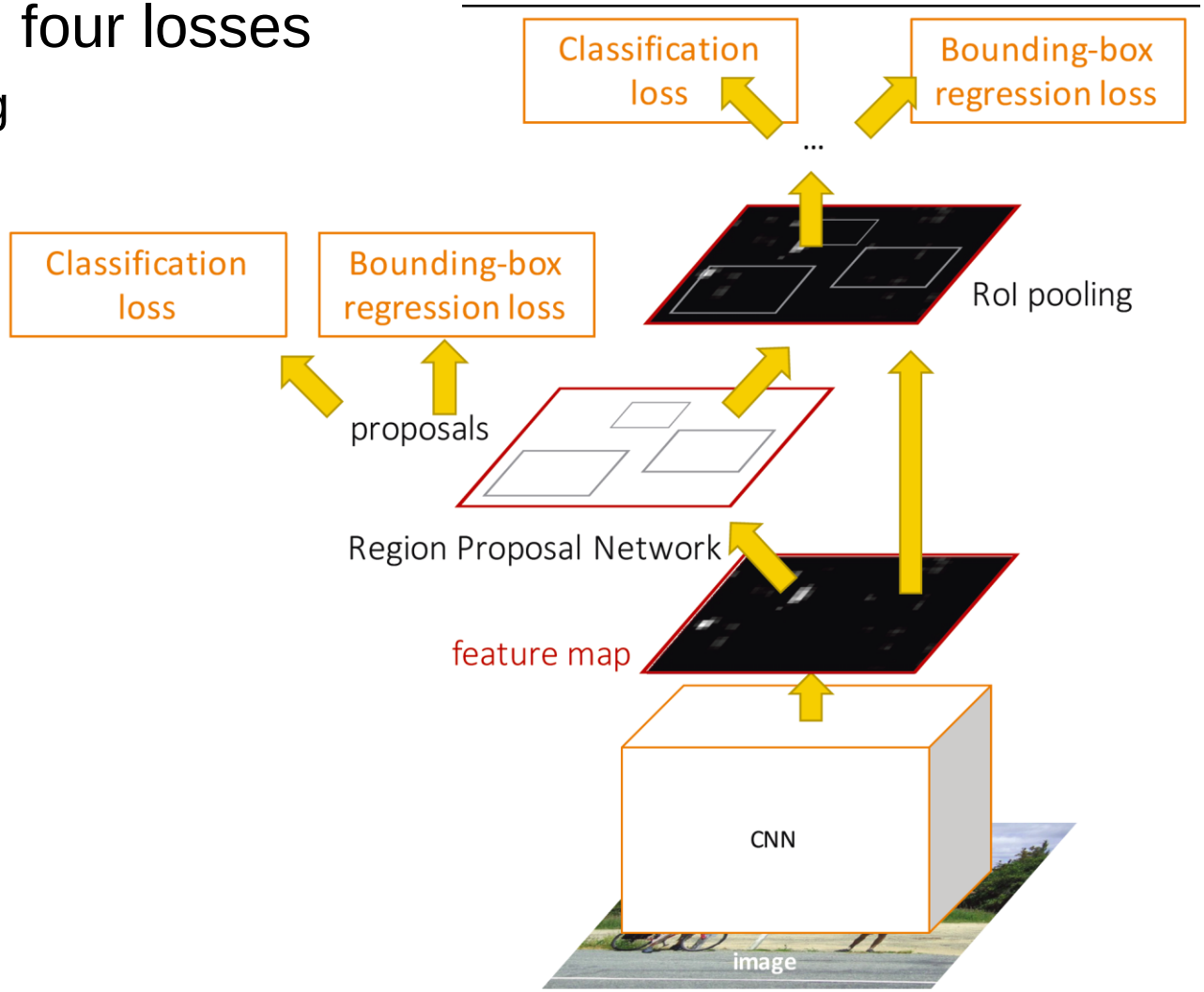
- Faster R-CNN = Fast R-CNN + RPN



Slide credit: Ross Girshick

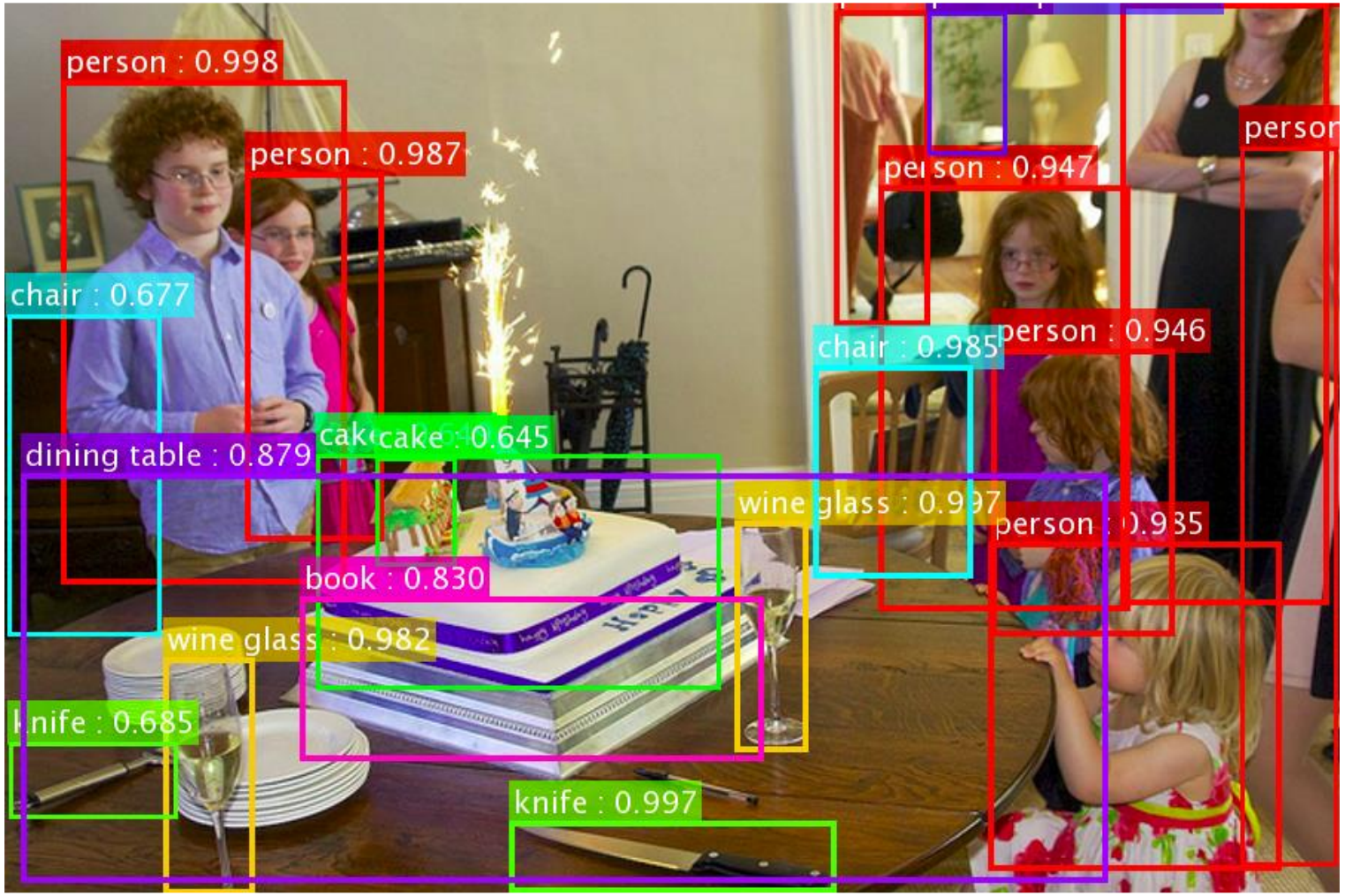
Faster R-CNN

- One network, four losses
  - Joint training



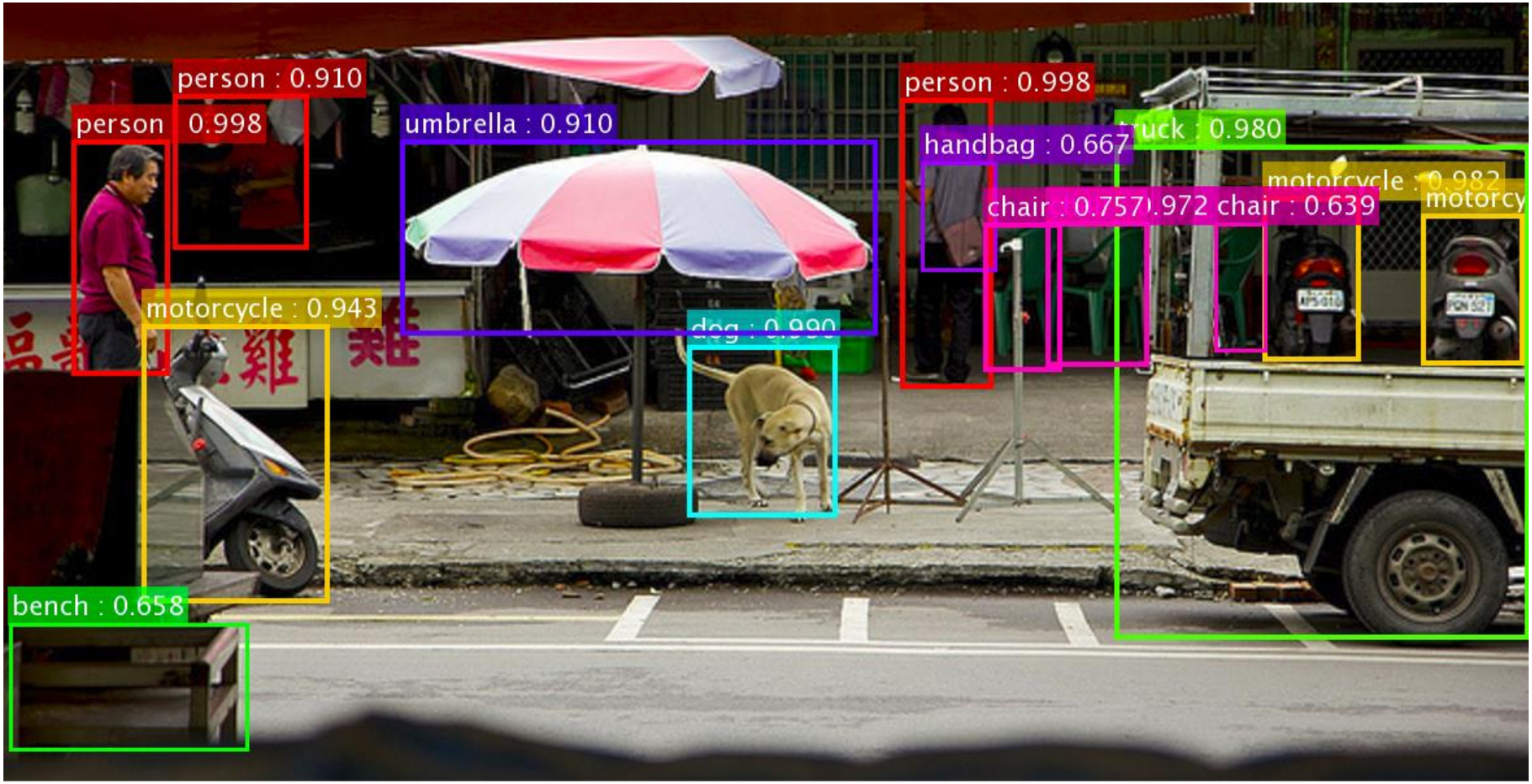
Slide credit: Ross Girshick

Faster R-CNN (based on ResNets)



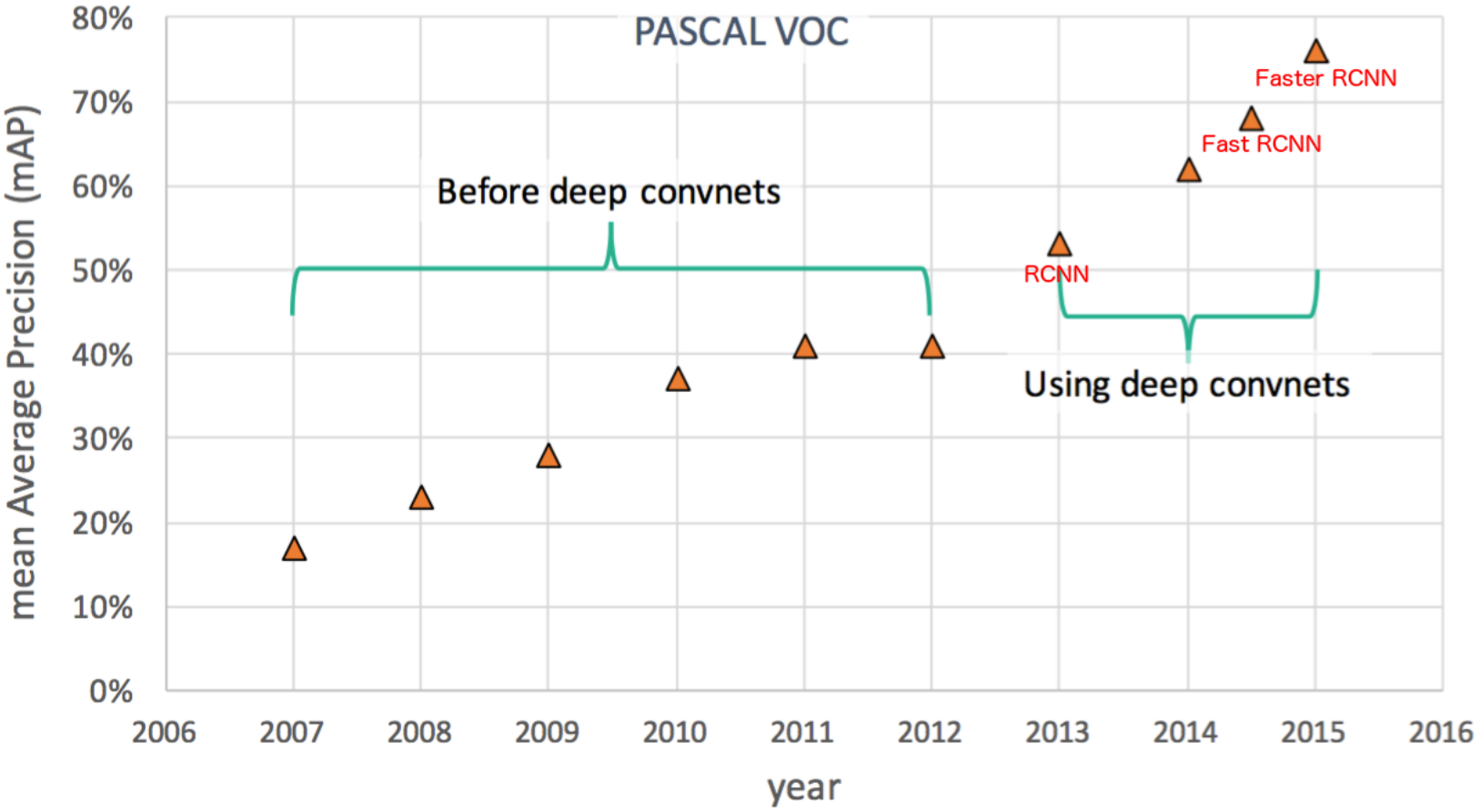
K. He, X. Zhang, S. Ren, J. Sun, Deep Residual Learning for Image Recognition, CVPR 2016.

Faster R-CNN (based on ResNets)



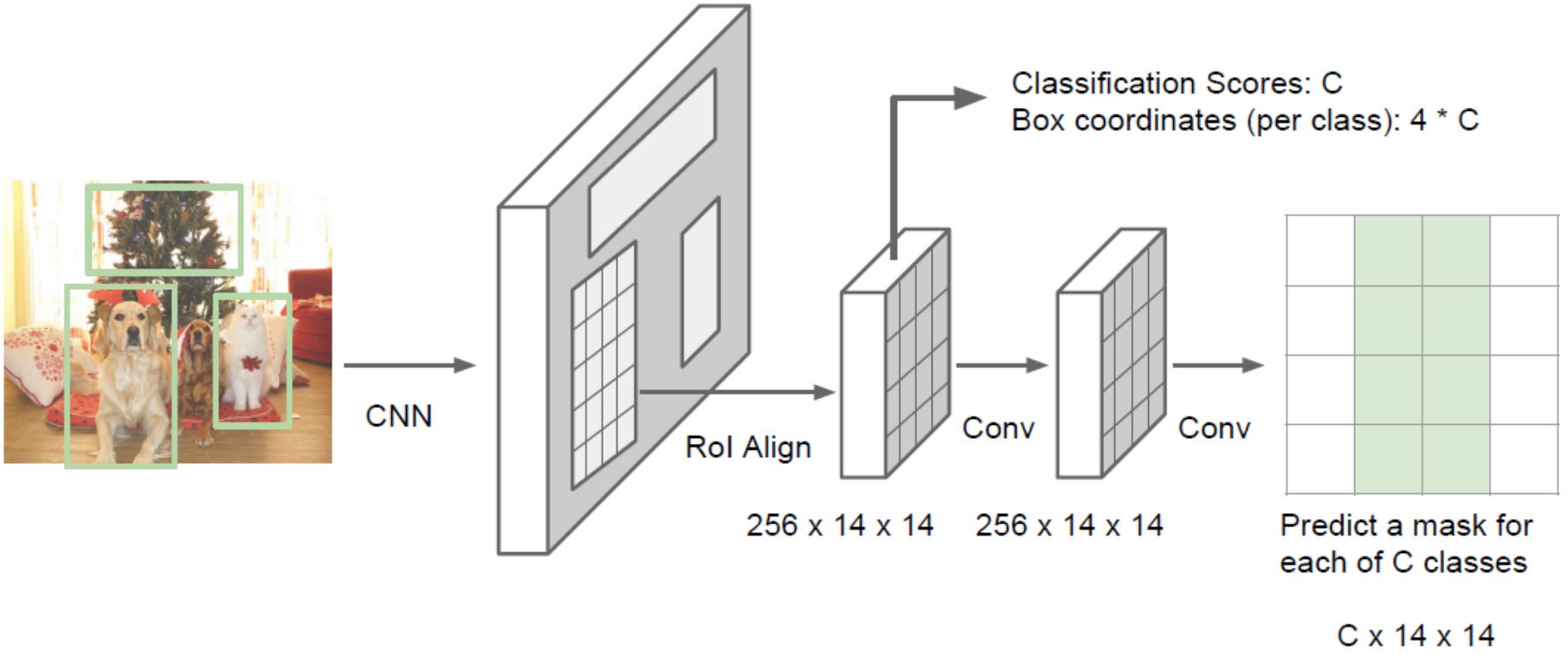
K. He, X. Zhang, S. Ren, J. Sun, Deep Residual Learning for Image Recognition, CVPR 2016.

Object Detection Performance



Slide credit: Ross Girshick

More Recent Version: Mask R-CNN

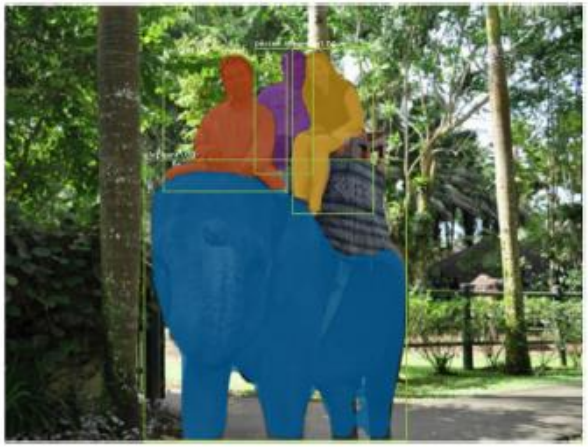
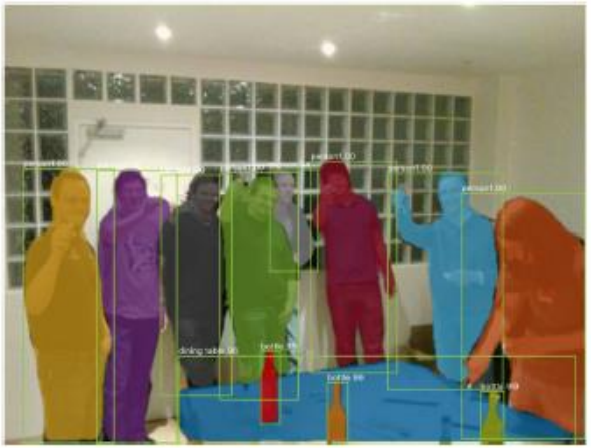


K. He, G. Gkioxari, P. Dollar, R. Girshick, Mask R-CNN, arXiv 1703.06870.

Slide credit: FeiFei Li

Mask R-CNN Results

- Detection + Instance segmentation

- Detection + Pose estimation

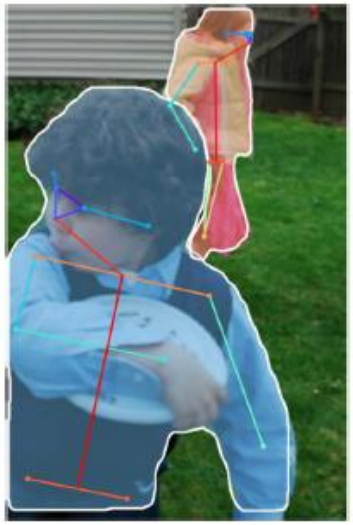
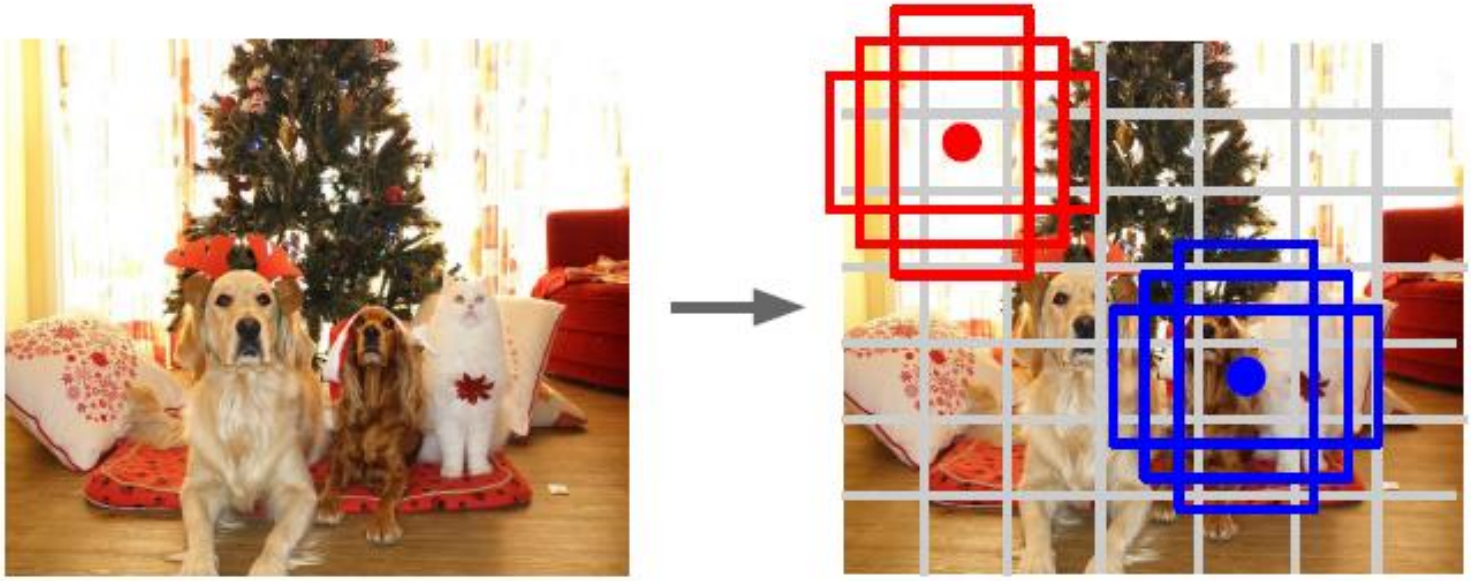



Figure credit: K. He, G. Gkioxari, P. Dollar, R. Girshick

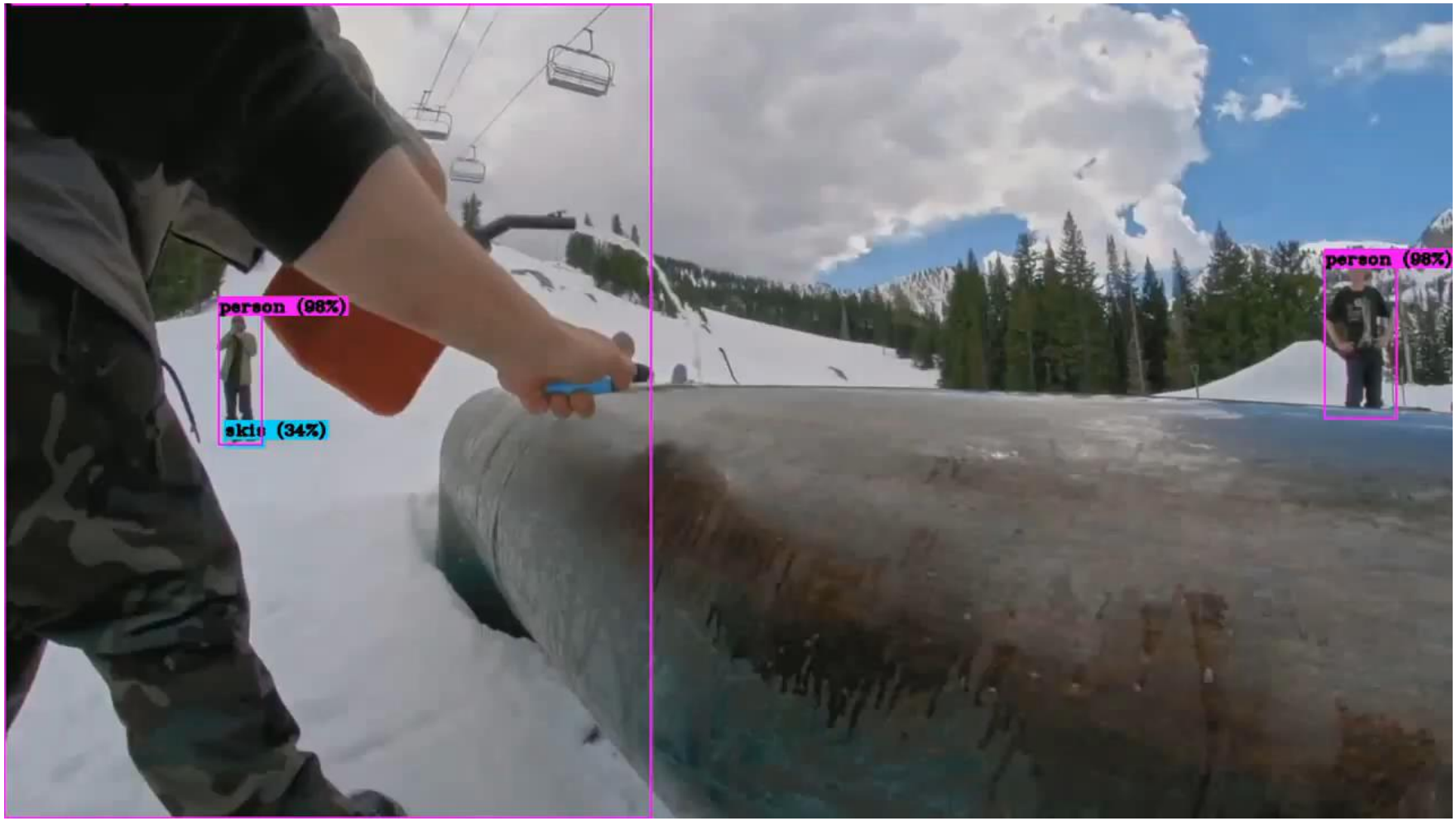
YOLO / SSD



- Idea: Directly go from image to detection scores
- Within each grid cell
  - Start from a set of anchor boxes
  - Regress from each of the B anchor boxes to a final box
  - Predict scores for each of C classes (including background)

Slide credit: FeiFei Li

YOLO-v4 Results



J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You Only Look Once: Unified, Real-Time Object Detection, CVPR 2016.

Summary

- Object Detection
  - Find a variable number of objects by classifying image regions
  - Before CNNs: dense multiscale sliding window (HoG, DPM)
- Region proposal based detectors
  - Idea: Avoid dense sliding window with region proposals
  - R-CNN: Selective Search + CNN classification / regression
  - Fast R-CNN: Swap order of convolutions and region extraction
  - Faster R-CNN: Compute region proposals within the network
  - Mask R-CNN: Detection + instance segmentation + pose estimation
- Anchor box based detectors
  - Idea: Perform detection in a single step using grid of anchor boxes
  - YOLO, YOLO-v2, YOLO-v3, …
  - SSD

Topics of This Lecture

- CNNs for Object Detection
  - The R-CNN Family
  - The YOLO Family
- CNNs for Segmentation
  - Fully Convolutional Networks (FCN)
  - Encoder-Decoder architecture
  - Transpose convolutions
  - Skip connections
- CNNs for Human Pose Estimation
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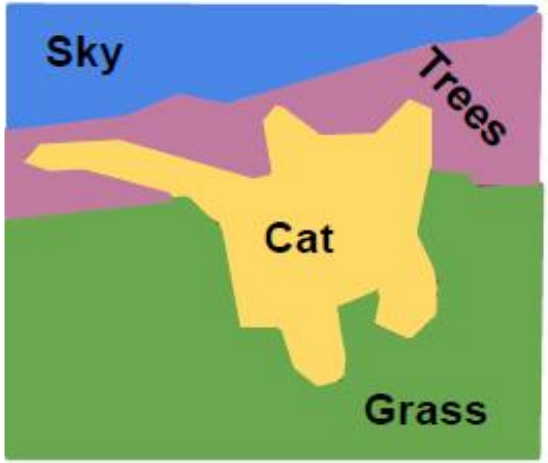
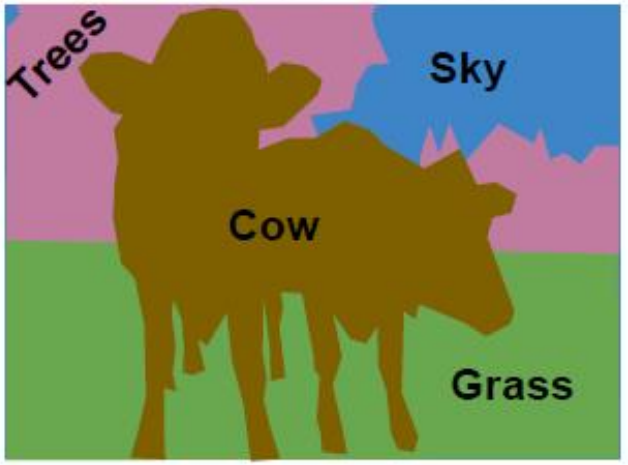
- CNNs for Matching
  - Siamese networks
  - Triplet loss

Semantic Segmentation

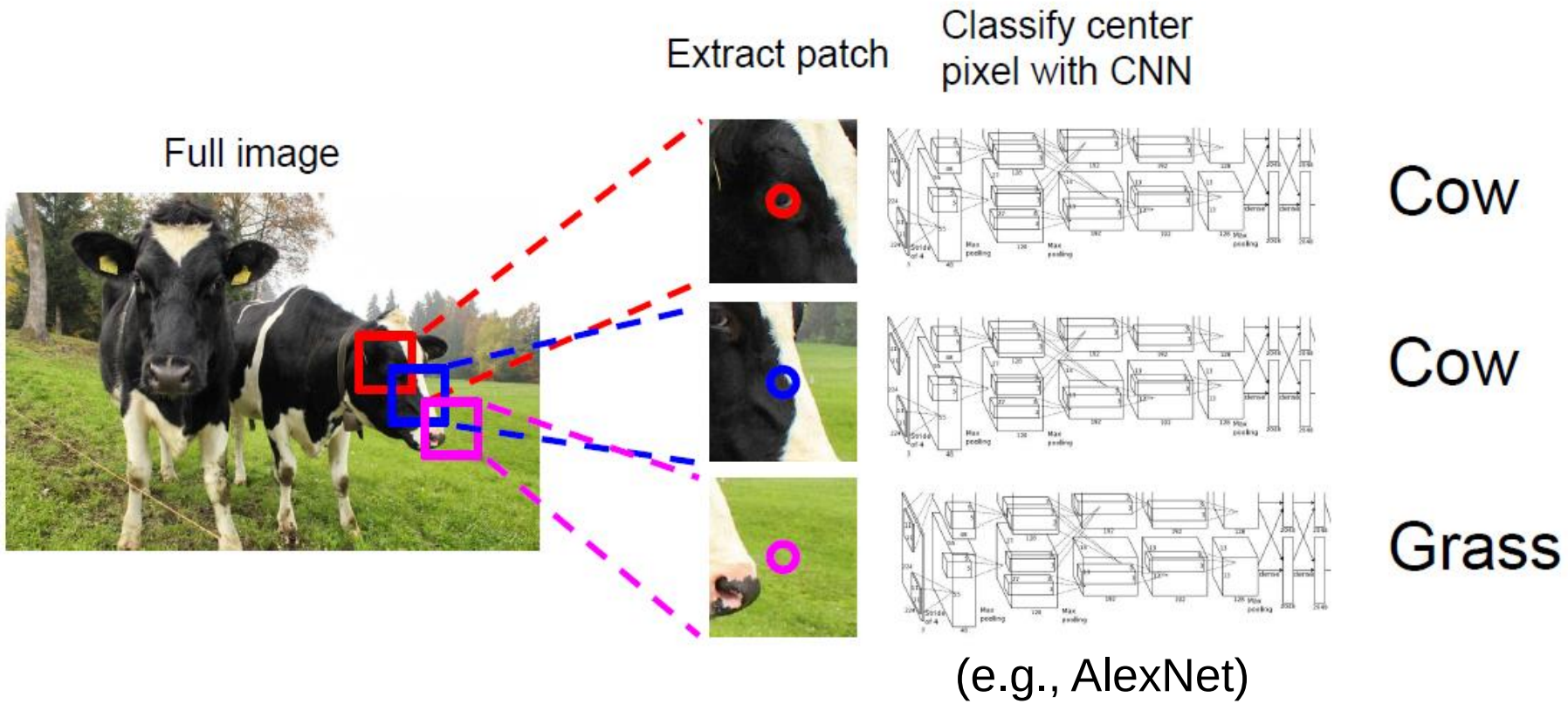
- Semantic Segmentation
  - Label each pixel in the image with a category label
  - Don’t differentiate instances, only care about pixels
- Instance segmentation
  - Also give an instance label per pixel





Slide adapted from FeiFei Li

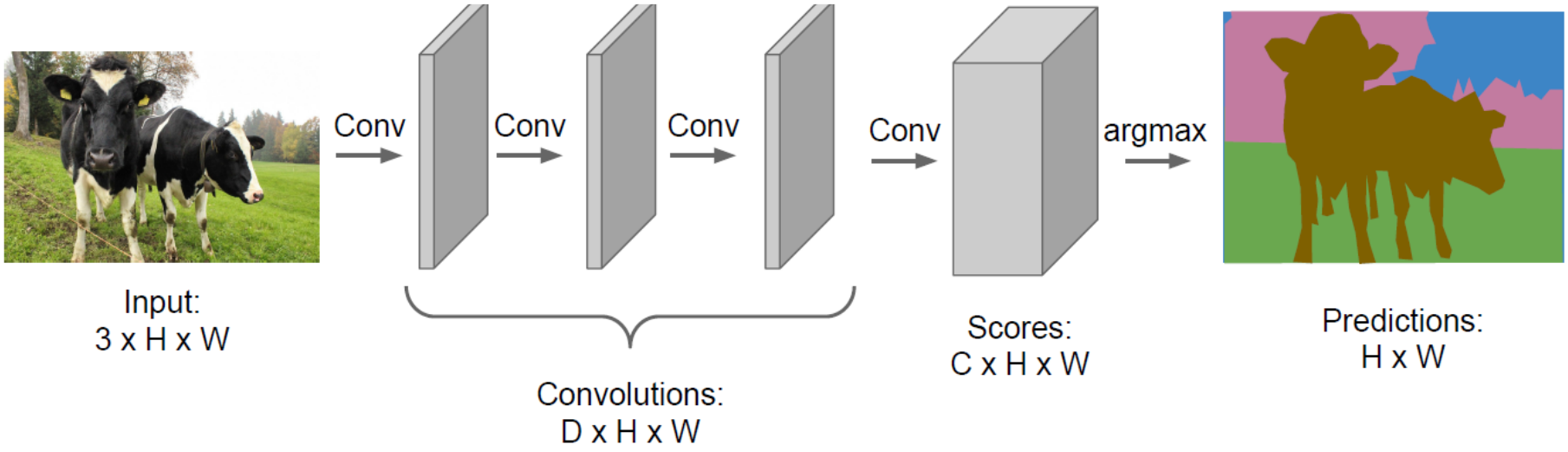
Segmentation Idea: Sliding Window



- Problem
  - Very inefficient
  - No reuse of features between shared patches

Slide adapted from FeiFei Li

Segmentation Idea: Fully-Convolutional Nets

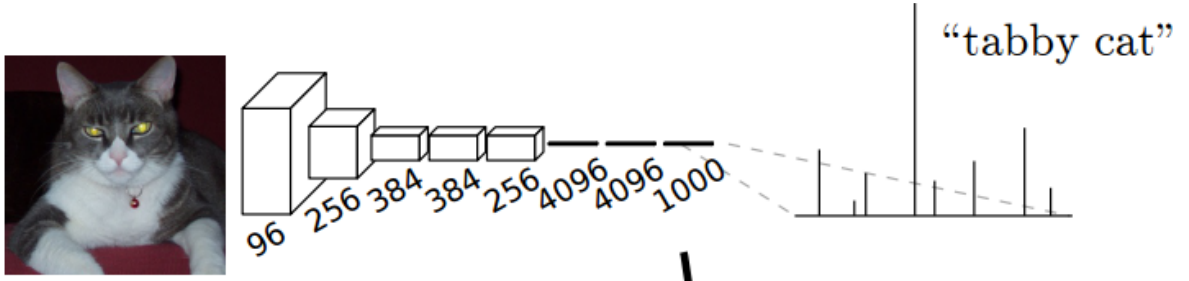


- Design a network as a sequence of convolutional layers
  - To make predictions for all pixels at once
  - Fully Convolutional Networks (FCNs)
    - All operations formulated as convolutions
    - Fully-connected layers become 1×1 convolutions
    - Advantage: can process arbitrarily sized images

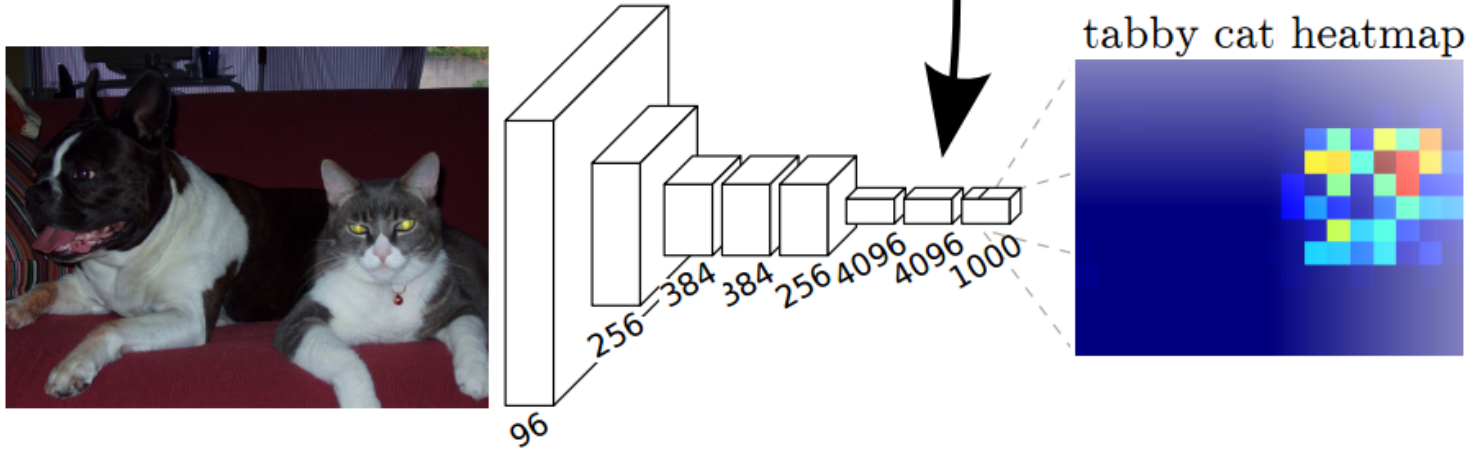
Slide adapted from FeiFei Li

CNNs vs. FCNs

- CNN



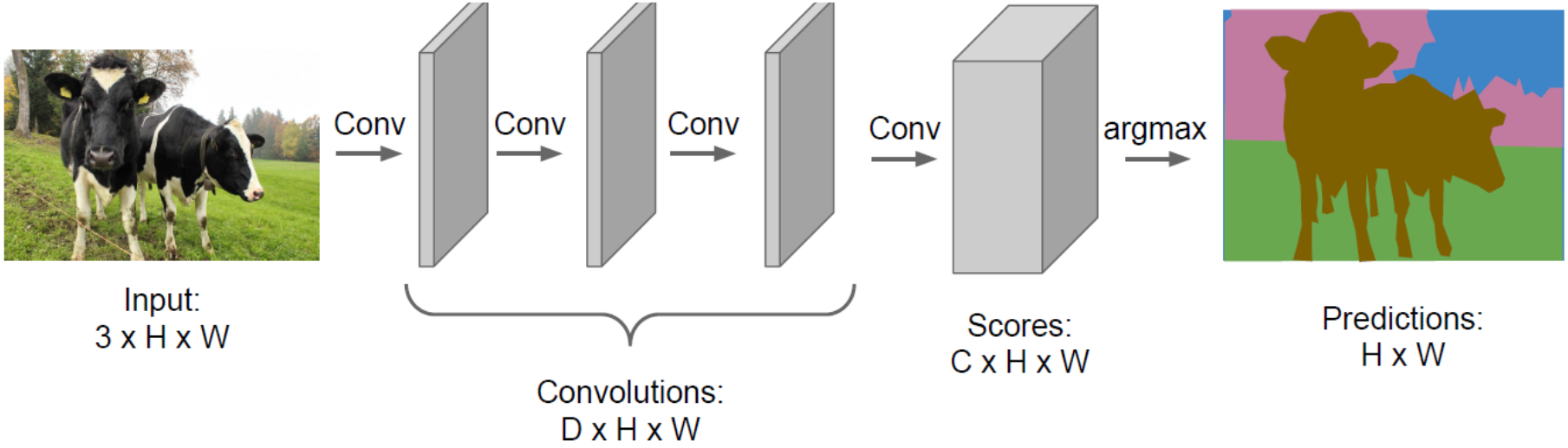
- FCN



- Intuition
  - Think of FCNs as performing a sliding-window classification, producing a heatmap of output scores for each class
  - But: more efficient, since computations are reused between windows

Image source: Long, Shelhamer, Darrell

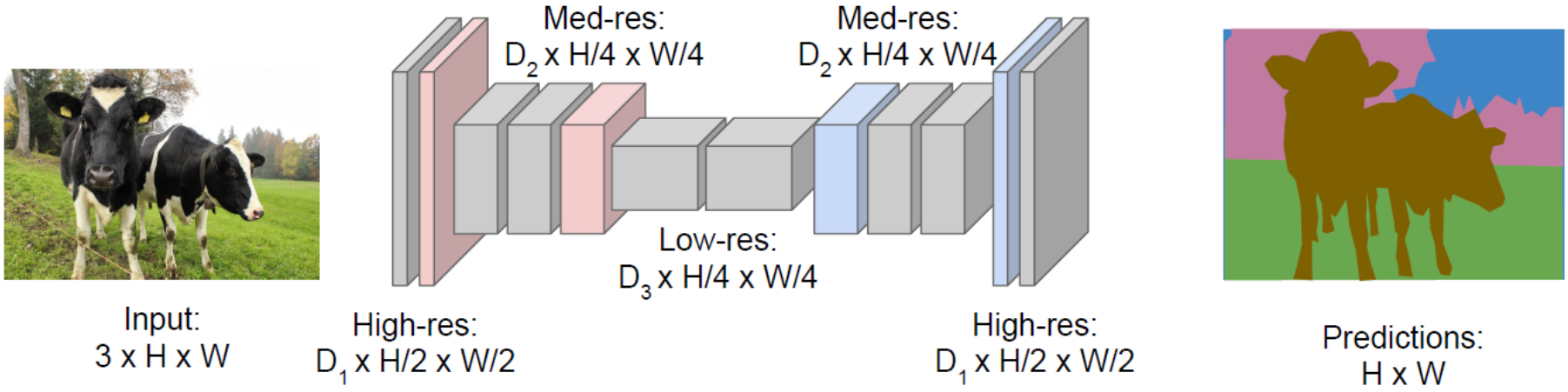
Segmentation Idea: Fully-Convolutional Nets



- Design a network as a sequence of convolutional layers
  - To make predictions for all pixels at once
- Problem
  - Convolutions at original image resolution will be very expensive!

Slide adapted from FeiFei Li

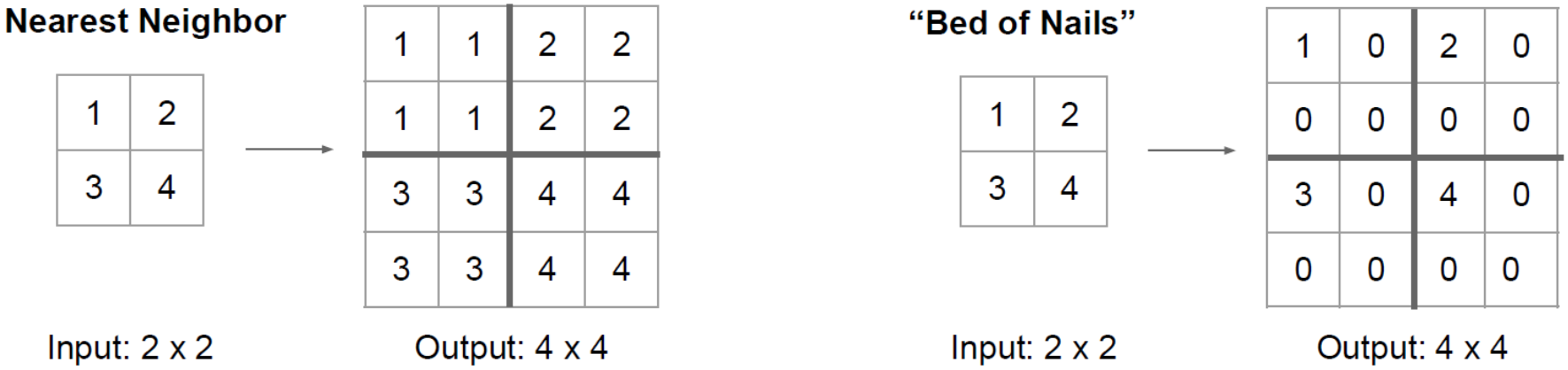
Segmentation Idea: Encoder-Decoder Nets



- Design a network as a sequence of convolutional layers
  - With downsampling and upsampling inside the network!
  - Downsampling
    - Pooling, strided convolution
  - Upsampling
    - ???

Slide credit: FeiFei Li

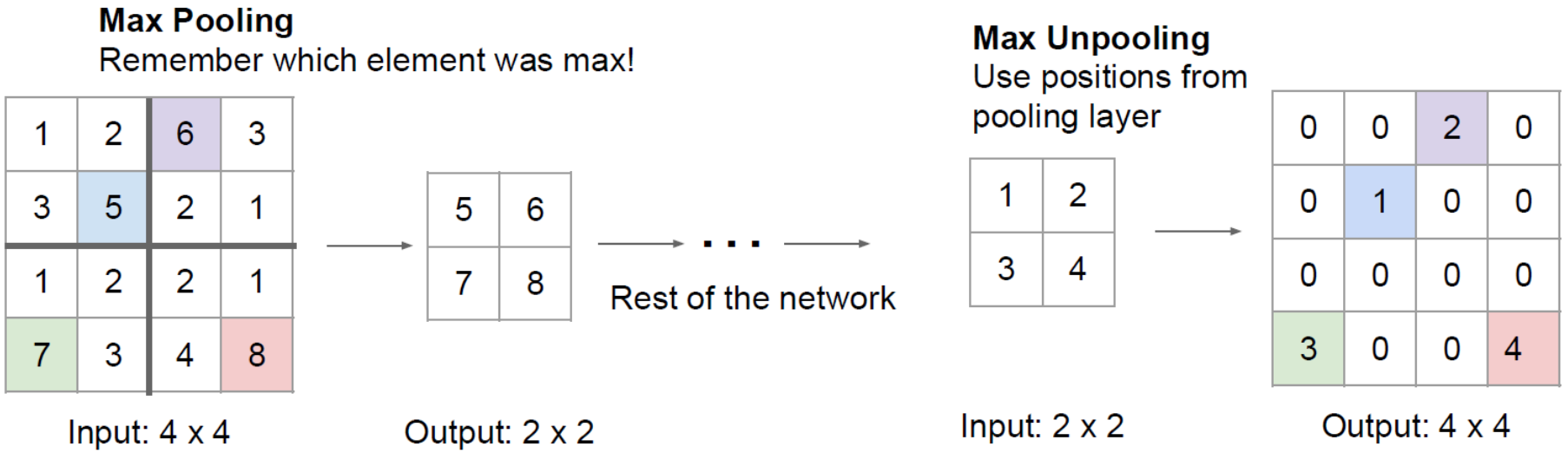
In-Network Upsampling: “Unpooling”



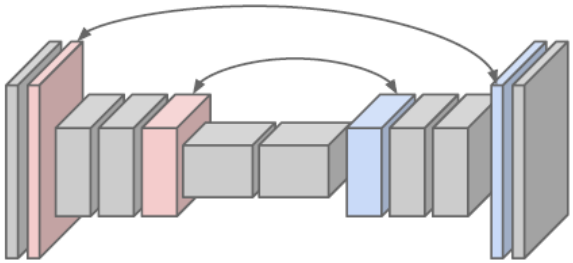
- Nearest-Neighbor
  - Simplest version
  - Problem: blocky output structure
- “Bed of Nails”
  - Preserve fine-grained structure of the output
  - Problem: fixed location for upsampled stimuli

Slide credit: FeiFei Li

In-Network Upsampling: "Max Unpooling"



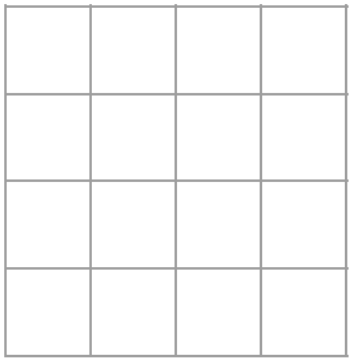
- Max Unpooling
  - Use corresponding pairs of downsampling and upsampling layers together
  - Remember which elements were max



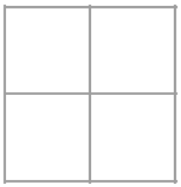
Slide credit: FeiFei Li

Learnable Upsampling: Transpose Convolution

- Recall: Normal convolution, stride 2, pad 1



Input: 4 x 4



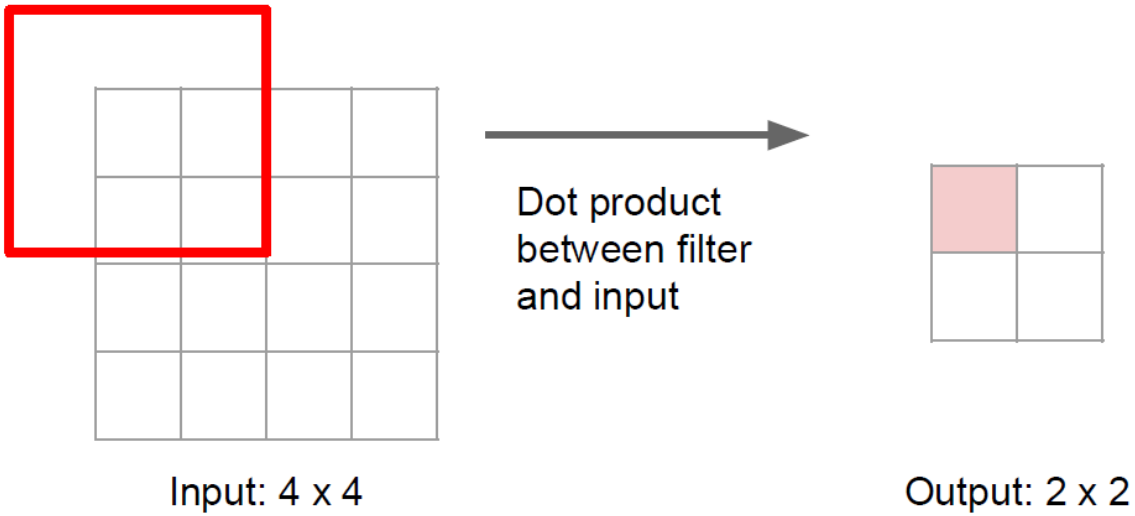
Output: 2 x 2

- Effect
  - Filter moves 2 pixels in the input for every one pixel in the output
  - Stride gives ration between movement in input and output

Slide credit: FeiFei Li

Learnable Upsampling: Transpose Convolution

- Recall: Normal convolution, stride 2 pad 1



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Learnable Upsampling: Transpose Convolution

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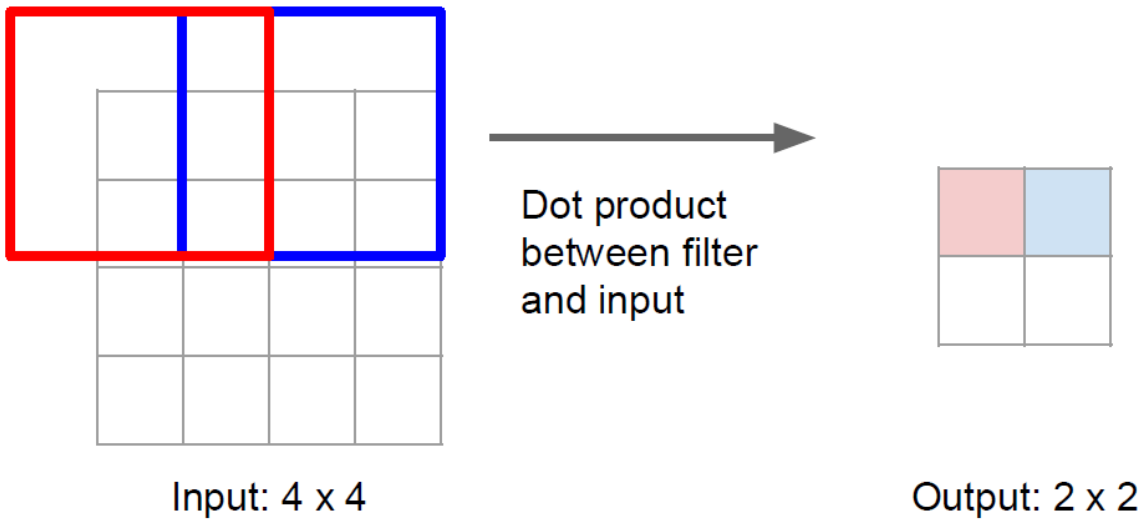
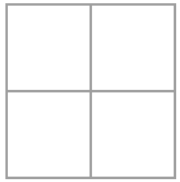


- Effect
  - Filter moves 2 pixels in the input for every one pixel in the output
  - Stride gives ration between movement in input and output

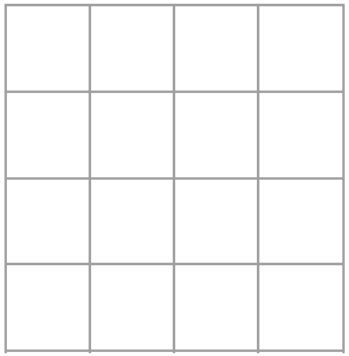
Slide credit: FeiFei Li

Learnable Upsampling: Transpose Convolution

- Now: 3x3 transpose convolution, stride 2 pad 1



Input: 2 x 2

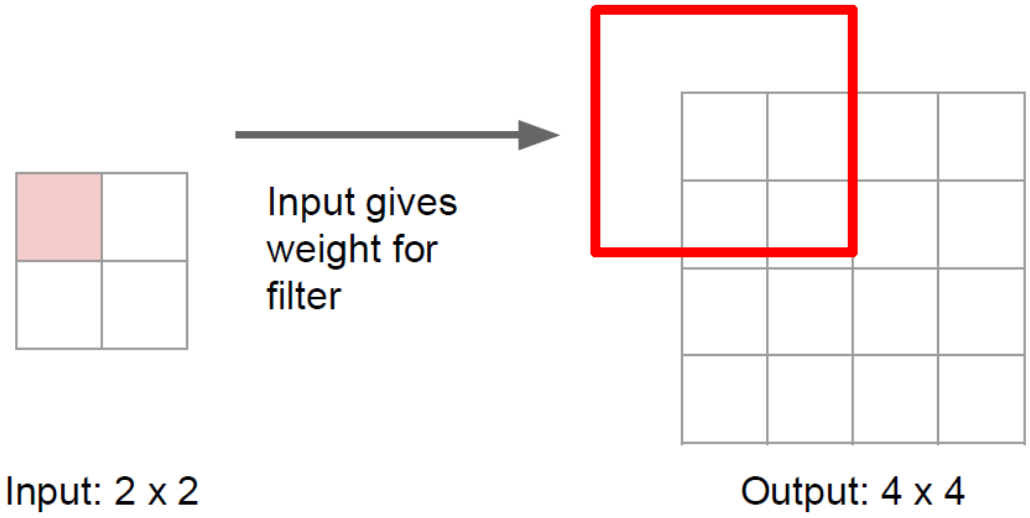


Output: 4 x 4

Slide credit: FeiFei Li

Learnable Upsampling: Transpose Convolution

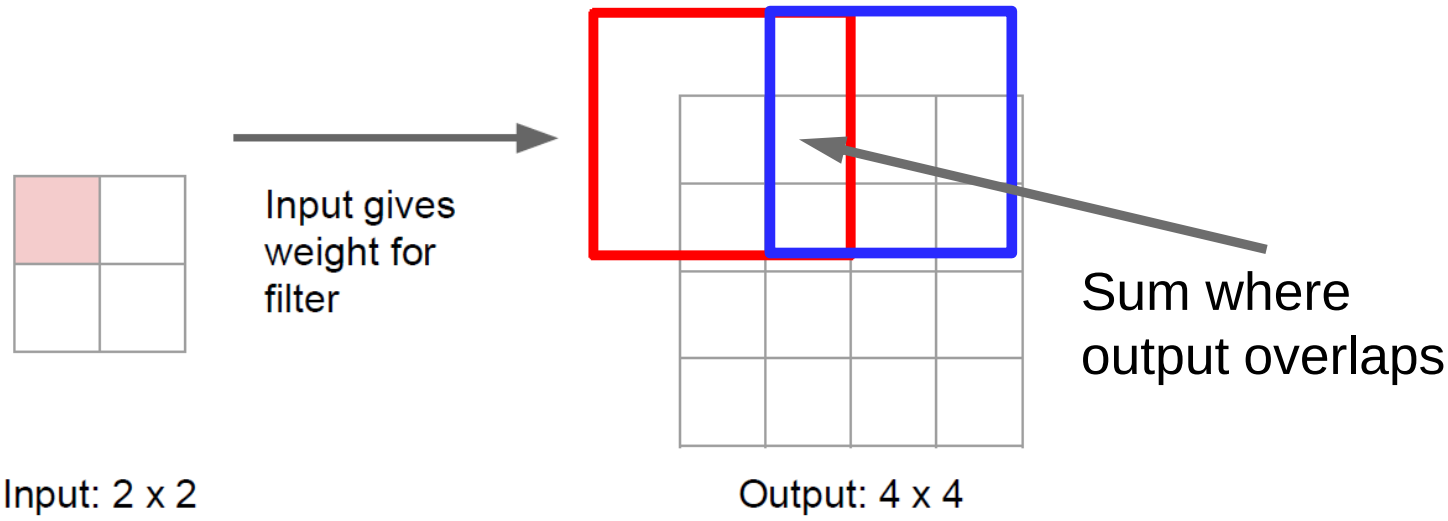
- Now: 3x3 transpose convolution, stride 2 pad 1



Slide credit: FeiFei Li

Learnable Upsampling: Transpose Convolution

- Now: 3x3 transpose convolution, stride 2 pad 1

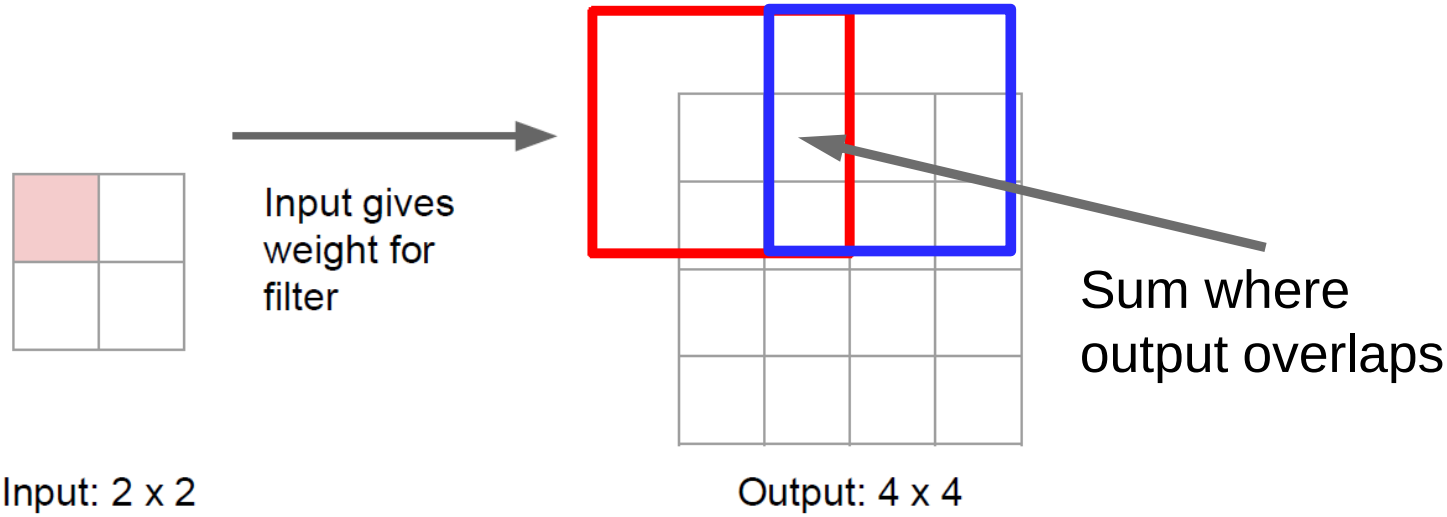


- Effect
  - Filter moves 2 pixels in the *output* for every one pixel in the *input*
  - Stride gives ration between movement in output and input

Slide credit: FeiFei Li

Learnable Upsampling: Transpose Convolution

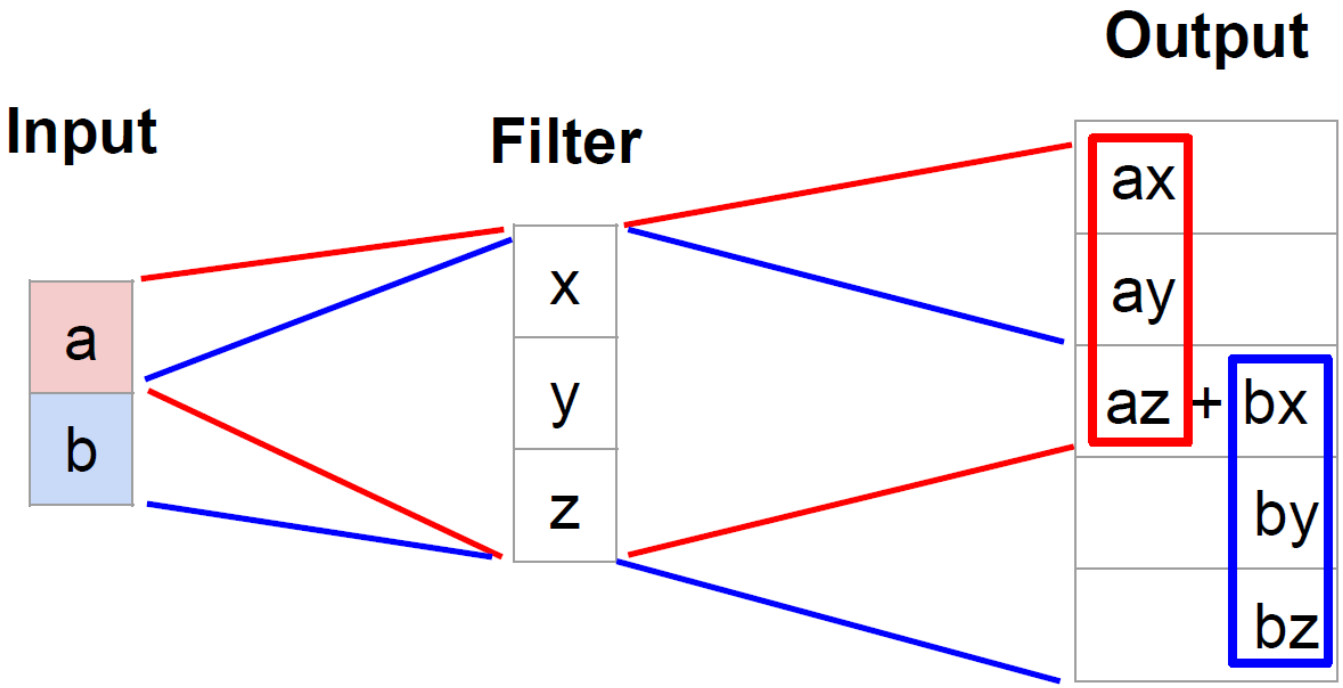
- Now: 3x3 transpose convolution, stride 2 pad 1



- Other names
  - Deconvolution (bad)
  - Upconvolution
  - Fractionally strided convolution
  - Backward strided convolution

Slide credit: FeiFei Li

Learnable Upsampling: 1D Example



- Observations
  - Output contains copies of the filter weighted by the input, summing overlaps in the output
  - Need to crop one pixel from output to make output exactly 2x input

Slide credit: FeiFei Li

Convolution as Matrix Multiplication (1D Example)

- Express convolution in terms of matrix multiplication
  - Example:
    - 1D conv
    - Kernel size = 3
    - Stride 1, padding = 1

$$\vec{x} * \vec{a} = X\vec{a}$$

$$\begin{bmatrix} x & y & z & 0 & 0 & 0 \\ 0 & x & y & z & 0 & 0 \\ 0 & 0 & x & y & z & 0 \\ 0 & 0 & 0 & x & y & z \end{bmatrix} \begin{bmatrix} 0 \\ a \\ b \\ c \\ d \\ 0 \end{bmatrix} = \begin{bmatrix} ay + bz \\ ax + by + cz \\ bx + cy + dz \\ cx + dy \end{bmatrix}$$

- Convolution transpose multiplies by the transpose of the same matrix
  - When stride = 1, convolution transpose is just a regular convolution (with different padding rules)

$$\vec{x} *^T \vec{a} = X^T \vec{a}$$

$$\begin{bmatrix} x & 0 & 0 & 0 \\ y & x & 0 & 0 \\ z & y & x & 0 \\ 0 & z & y & x \\ 0 & 0 & z & y \\ 0 & 0 & 0 & z \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} ax \\ ay + bx \\ az + by + cx \\ bz + cy + dx \\ cz + dy \\ dz \end{bmatrix}$$

Slide credit: FeiFei Li

Convolution as Matrix Multiplication (1D Example)

- Express convolution in terms of matrix multiplication
  - Example:
    - 1D conv
    - Kernel size = 3
    - Stride 2, padding = 1

$$\vec{x} * \vec{a} = X\vec{a}$$

$$\begin{bmatrix} x & y & z & 0 & 0 & 0 \\ 0 & 0 & x & y & z & 0 \end{bmatrix} \begin{bmatrix} 0 \\ a \\ b \\ c \\ d \\ 0 \end{bmatrix} = \begin{bmatrix} ay + bz \\ bx + cy + dz \end{bmatrix}$$

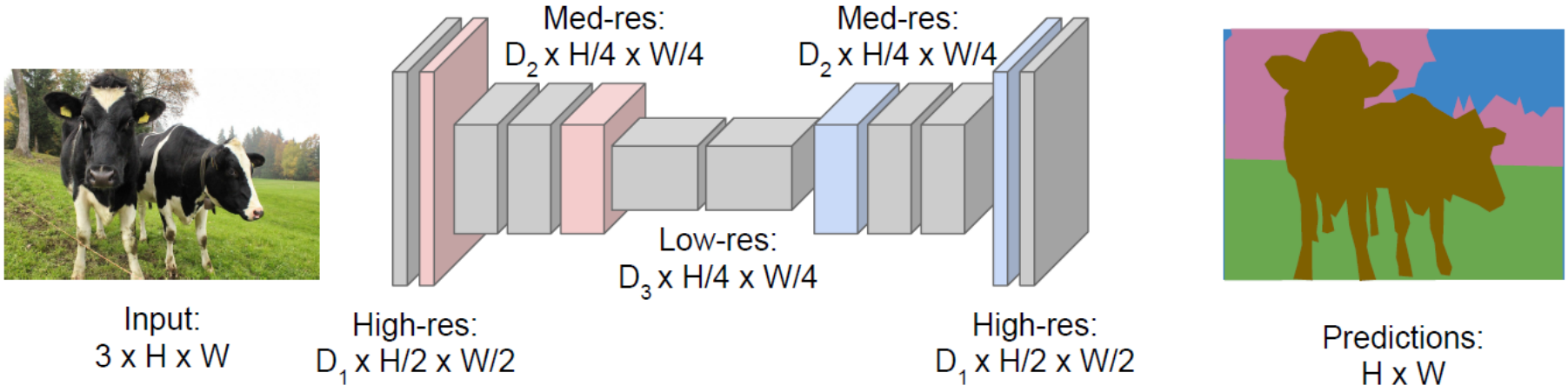
- Convolution transpose multiplies by the transpose of the same matrix
  - When stride > 1, convolution transpose is **no longer a normal convolution!**

$$\vec{x} *^T \vec{a} = X^T \vec{a}$$

$$\begin{bmatrix} x & 0 \\ y & 0 \\ z & x \\ 0 & y \\ 0 & z \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ax \\ ay \\ az + bx \\ by \\ bz \\ 0 \end{bmatrix}$$

Slide credit: FeiFei Li

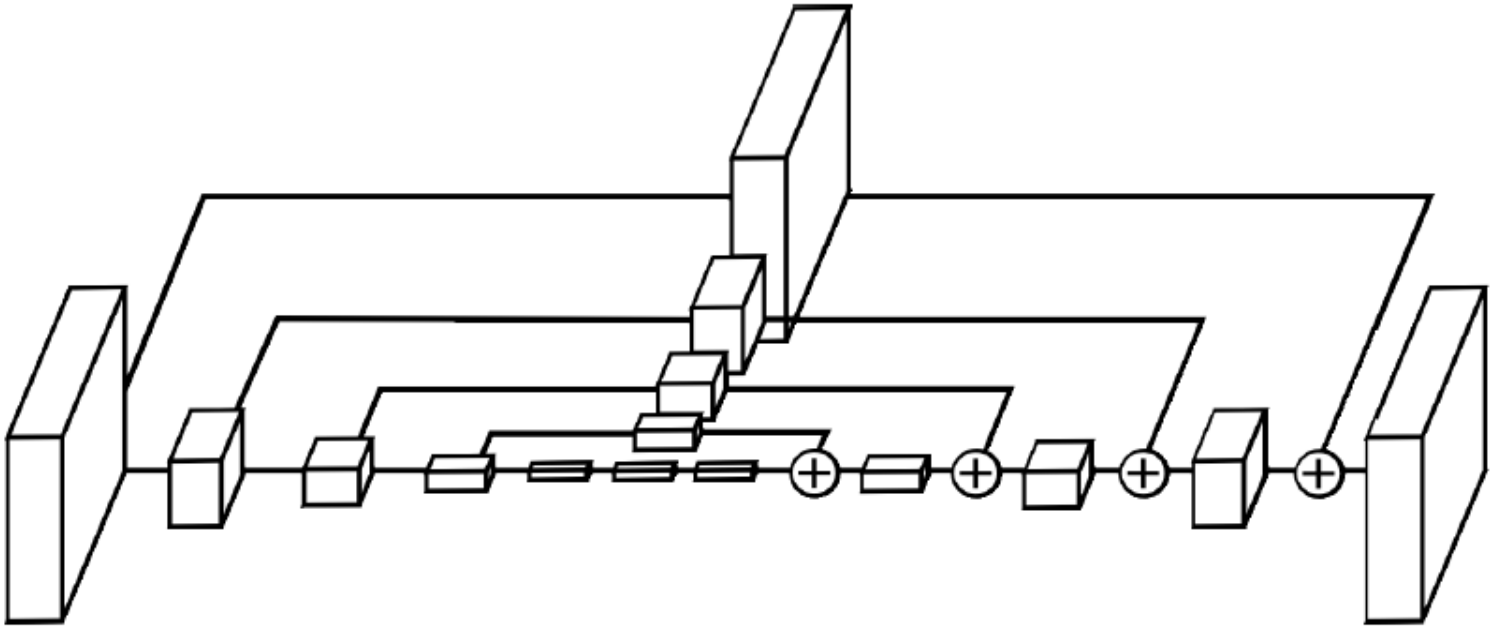
Segmentation Idea: Encoder-Decoder Nets



- Design a network as a sequence of convolutional layers
  - With downsampling and upsampling inside the network!
  - Downsampling
    - Pooling, strided convolution
  - Upsampling
    - Unpooling or strided transpose convolution

Slide credit: FeiFei Li

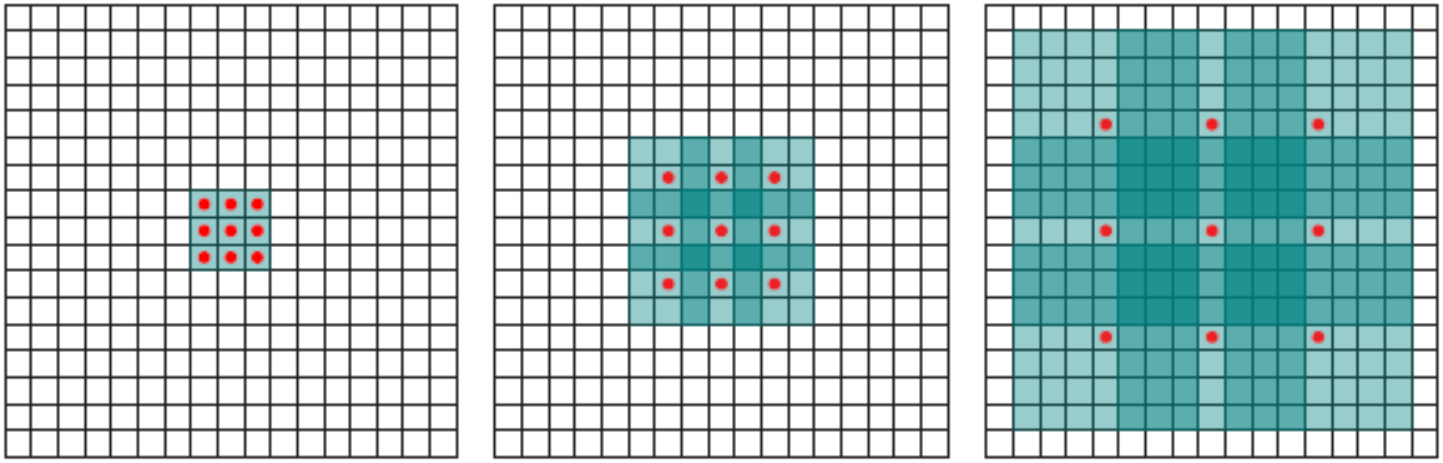
Extension: Skip Connections



- Encoder-Decoder Architecture with skip connections
  - Problem: downsampling loses high-resolution information
  - Use skip connections to preserve this higher-resolution information

Image source: Newell et al.

Extension: Dilated Convolutions



- Also known as: à trous convolutions or atrous convolutions
  - Simple way to increase the receptive field size without introducing additional parameters (filter size stays the same)

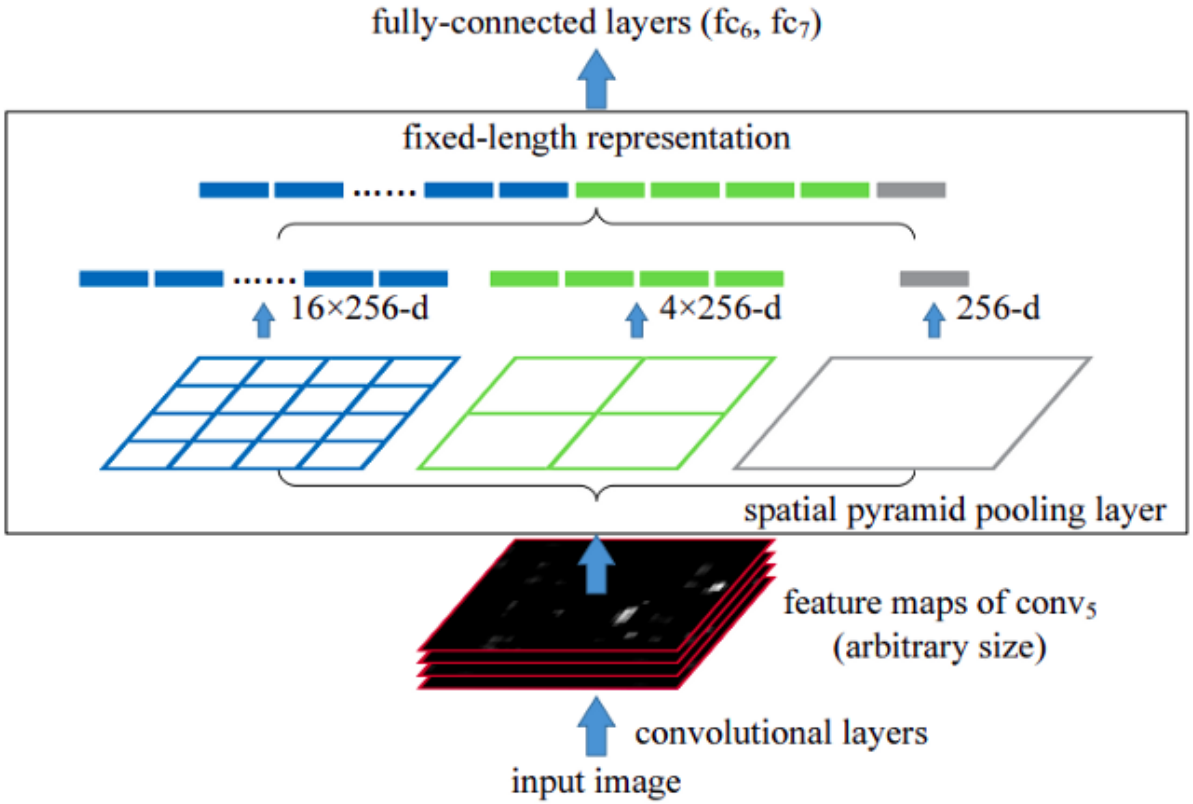
$$(F \star_l k)(\mathbf{p}) = \sum_{\mathbf{s}+l\mathbf{t}=\mathbf{p}} F(\mathbf{s})\, k(\mathbf{t}) \qquad \text{with dilation factor } l$$

  - Technique plays a key role in the *algorithme à trous* for wavelet decomposition [Holschneider et al., 1987]

Fisher Yu, Vladlen Koltun, Multi-Scale Context Aggregation by Dilated Convolutions, ICLR 2016

Image source: [Yu & Koltun, ICLR'16]

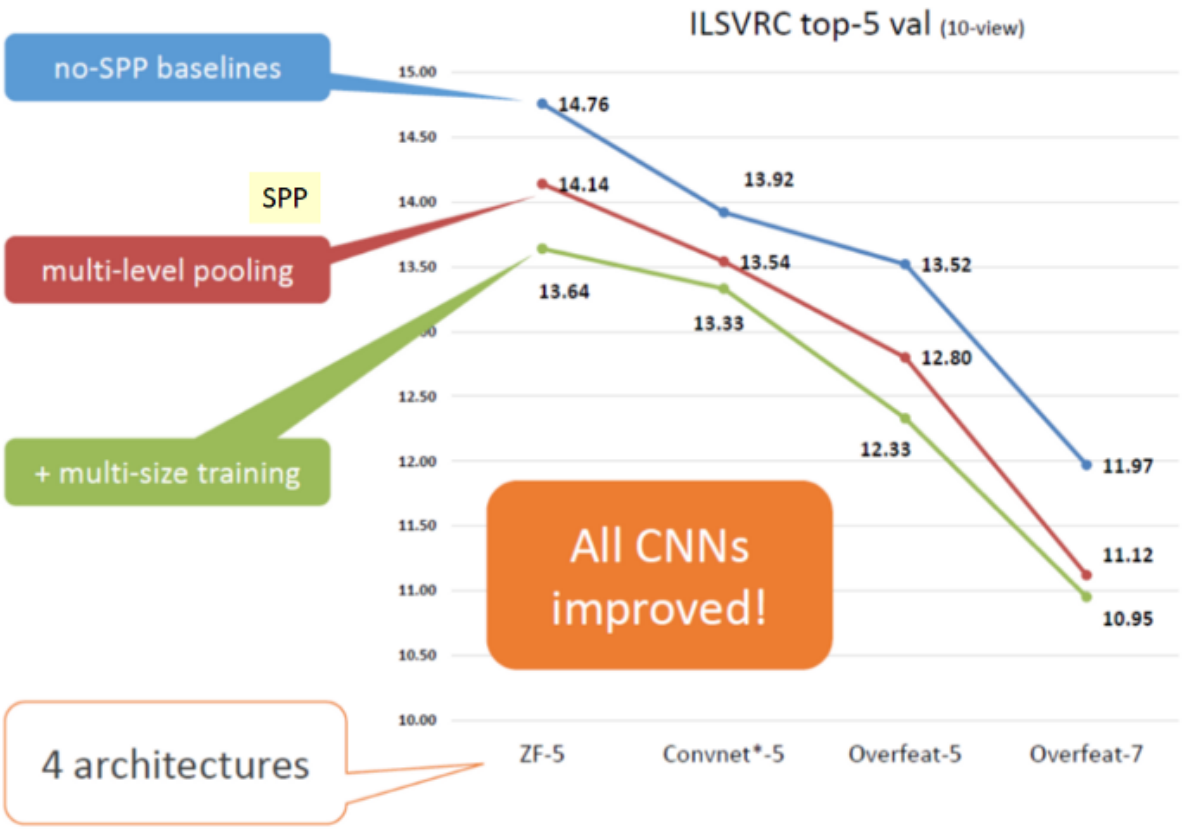
Extension: Spatial Pyramid Pooling (SPP)



- Multi-scale feature representation
  - Concatenated to a fixed-length vector

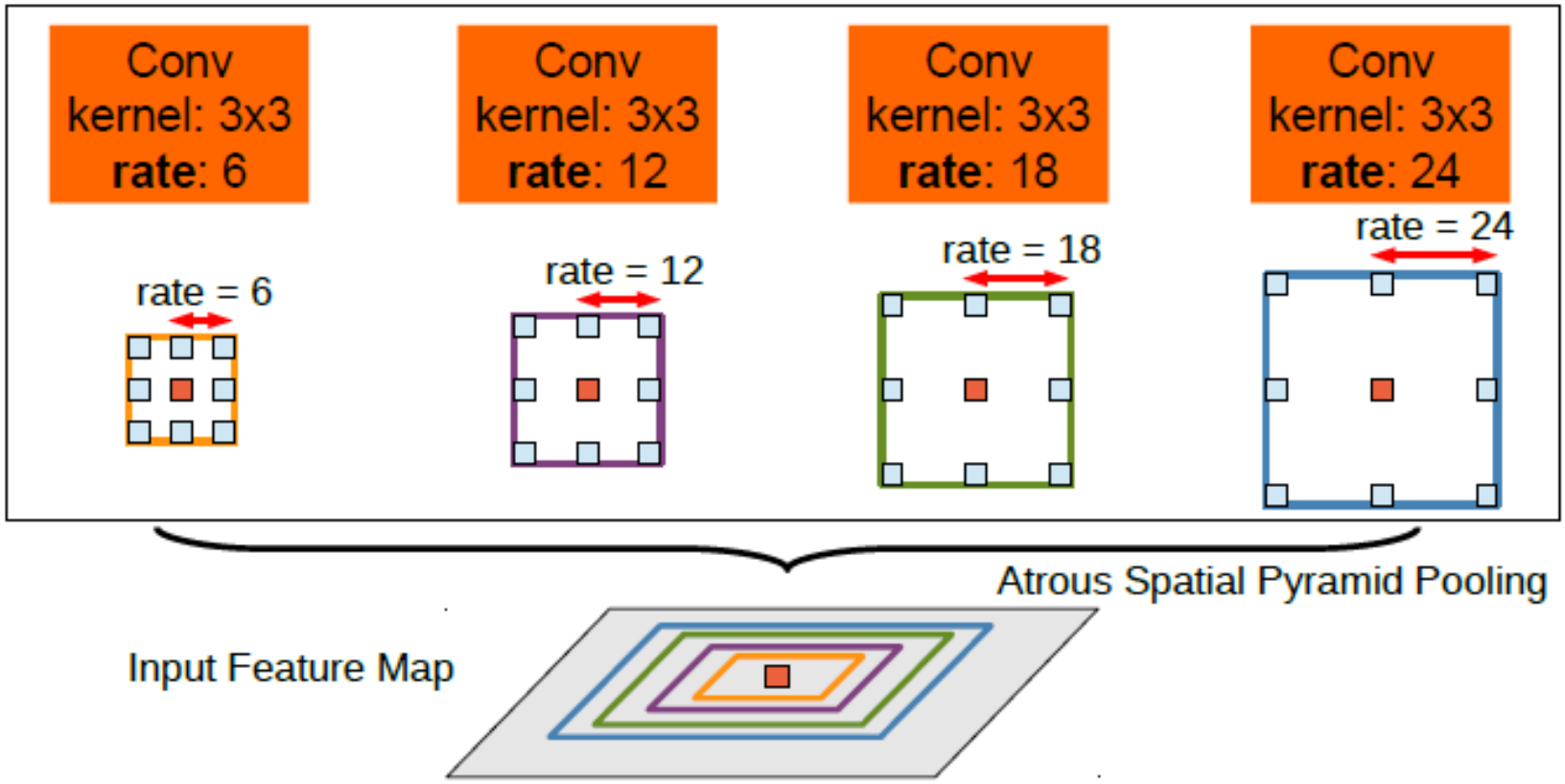
K. He, X. Zhang, S. Ren, J. Sun, Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition, ECCV 2014.

Spatial Pyramid Pooling (SPP)



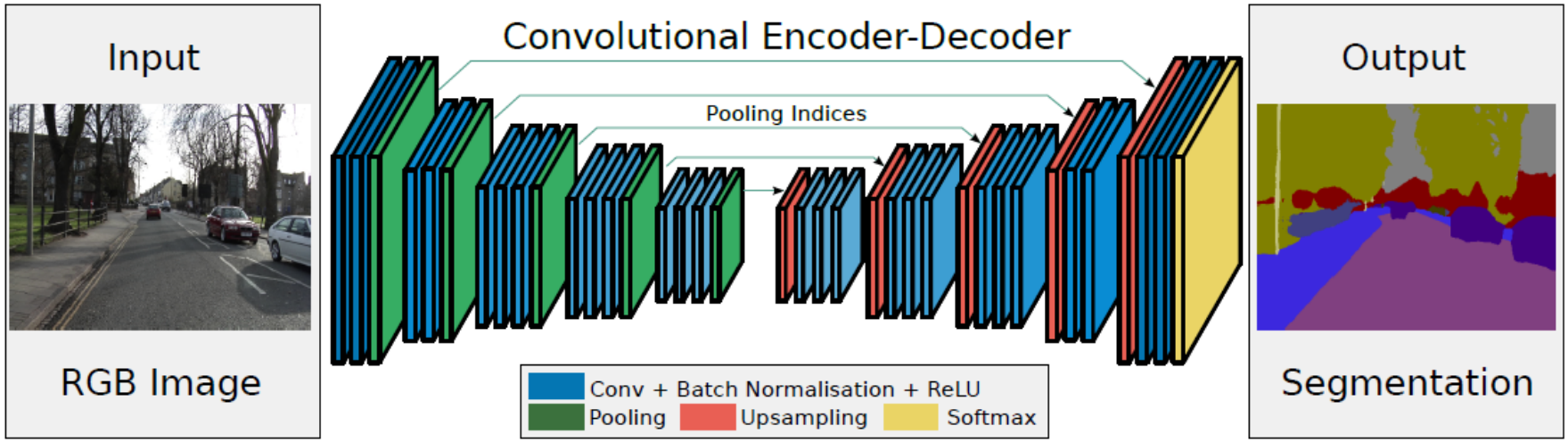
- Effect
  - Improved recognition performance for variety of architectures
  - SPP modules are used in many current architectures

Atrous Spatial Pyramid Pooling (ASPP)



- Extension of the SPP concept
  - Using dilated convolutions instead of regular convolutions
  - Effect: increased receptive field without extra computation

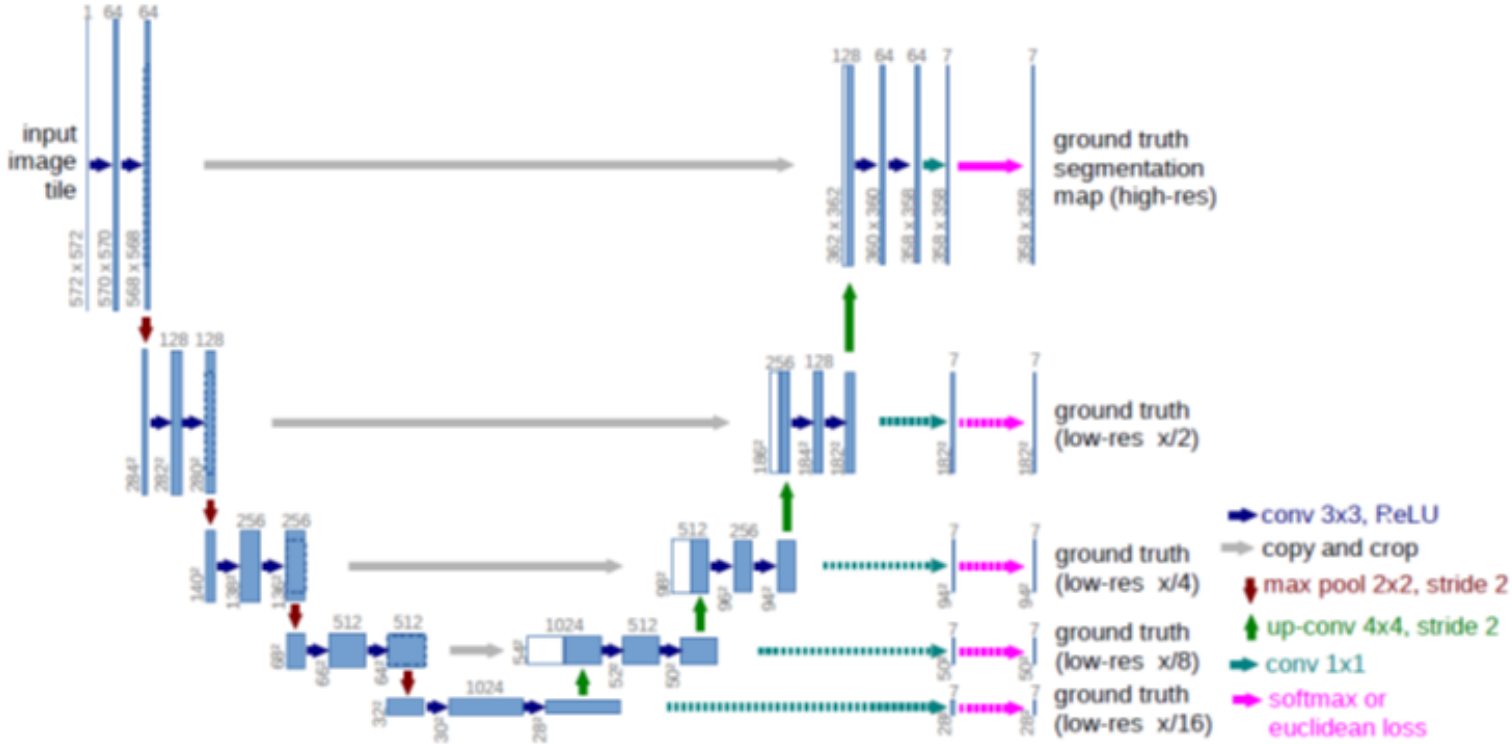
Example: SegNet



- SegNet
  - Encoder-Decoder architecture with skip connections
  - Encoder based on VGG-16
  - Decoder using Max Unpooling
  - Output with K-class Softmax classification

V. Badrinarayanan, A. Kendall, R. Cipolla, SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation, arXiv 1511.00561, IEEE Trans. PAMI 2017.

Example: U-Net

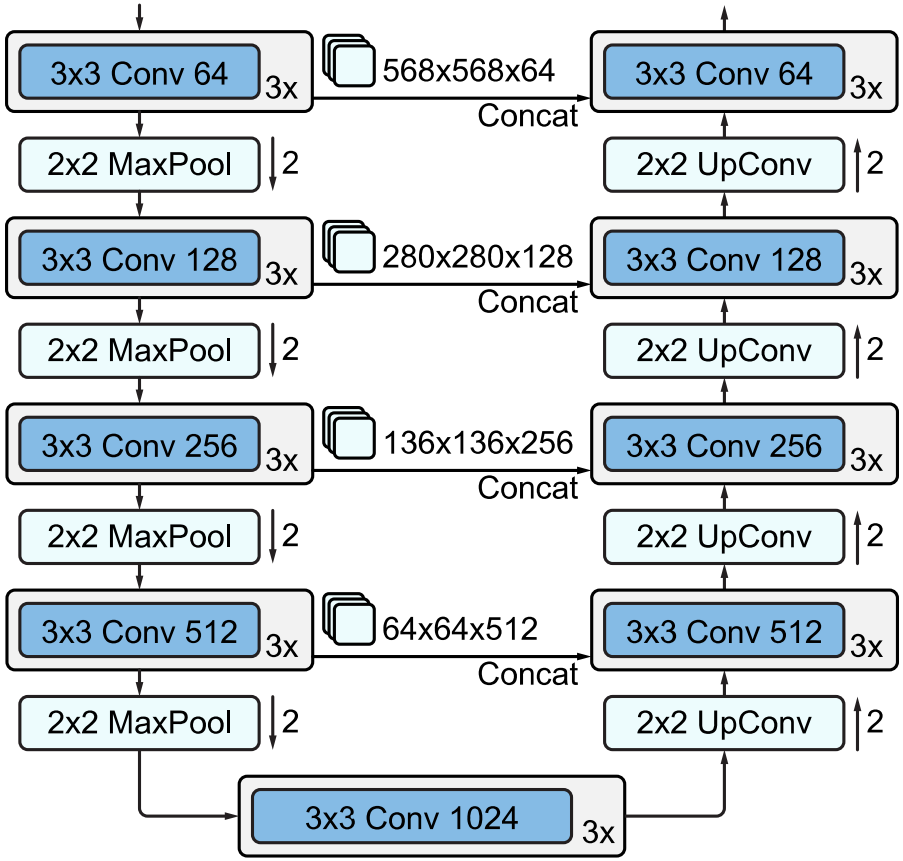


- U-Net
  - Similar idea, popular in biomedical image processing
  - Encoder-Decoder architecture with skip connections

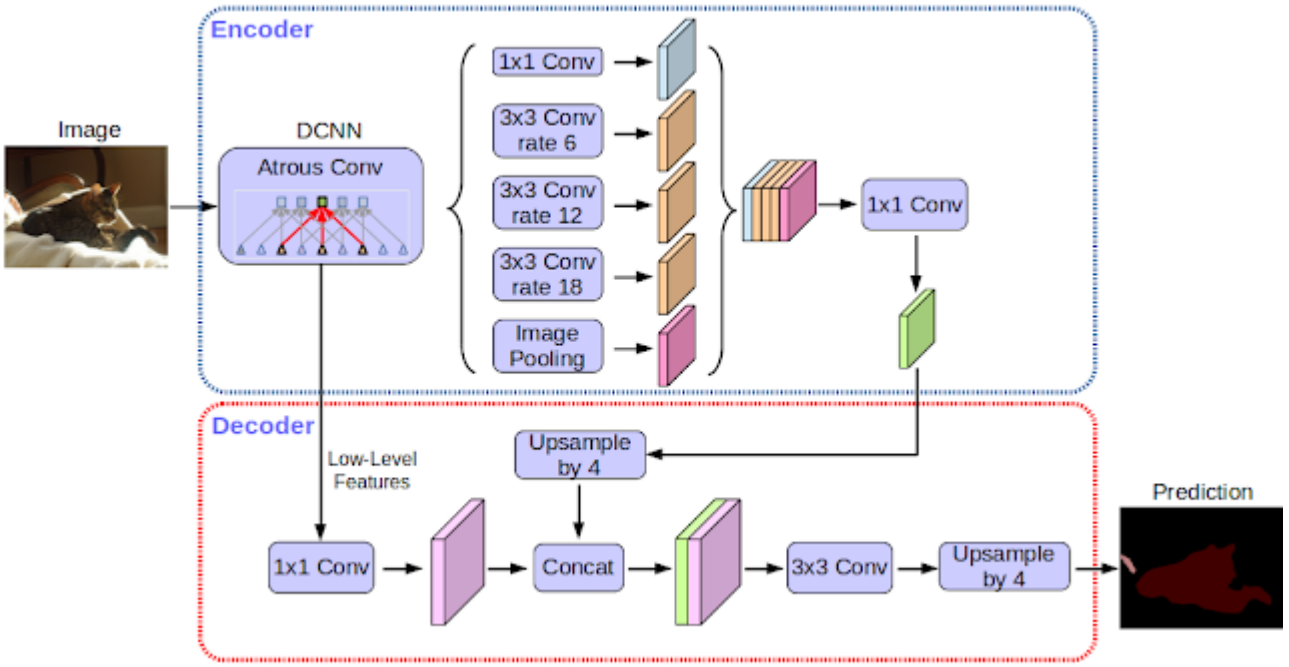
O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional Networks for Biomedical Image Segmentation, MICCAI 2015

Example: U-Net

- Encoder-Decoder structure
  - Encoder uses max pooling
  - Decoder uses transpose convolutions
  - Skip connections
    - Preserve high-resolution information



Example: DeepLabv3+



- Architecture designed for semantic segmentation
  - Uses atrous spatial pyramid pooling (ASPP) in the encoder
  - Simple decoder module
  - Depth-wise separable convolutions for efficiency

L-C. Chen, G. Papandreou, F. Schroff, H. Adam, DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolutions, and Fully Connected CRFs, IEEE Trans. PAMI, 2017

Semantic Segmentation

- Recent results
  - Based on an extension of ResNets for high-resolution segmentation

[Pohlen, Hermans, Mathias, Leibe, CVPR 2017]

Topics of This Lecture

- CNNs for Object Detection
  - The R-CNN Family
  - The YOLO Family
- CNNs for Segmentation
  - Fully Convolutional Networks (FCN)
  - Encoder-Decoder architecture
  - Transpose convolutions
  - Skip connections
- CNNs for Human Pose Estimation
  - 2D Body pose estimation
  - 3D Body pose estimation
- CNNs for Matching
  - Siamese networks
  - Triplet loss

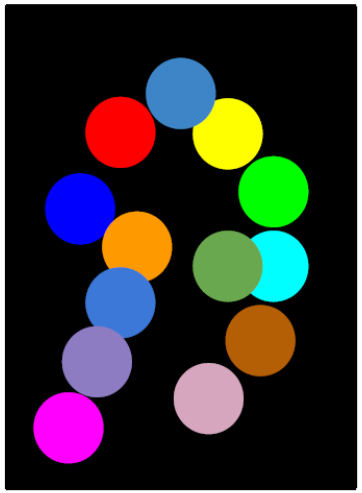
CNNs for Human Pose Estimation

- Input data

Image

Keypoints

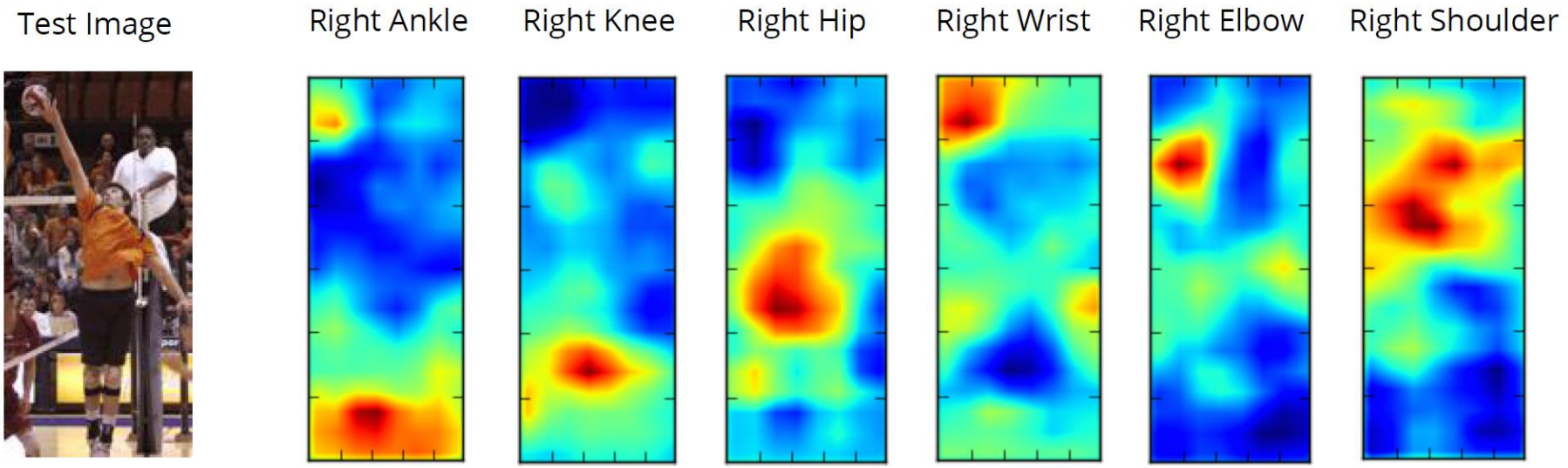
Labels



- Task setup
  - Annotate images with keypoints for skeleton joints
  - Define a target disk around each keypoint with radius r
  - Set the ground-truth label to 1 within each such disk
  - Infer heatmaps for the joints as in semantic segmentation

Slide adapted from Georgia Gkioxari

Heat Map Predictions from FCN



Slide adapted from Georgia Gkioxari

Example Results: Human Pose Estimation

[Rafi, Gall, Leibe, BMVC 2016]

Example Results: OpenPose

OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields

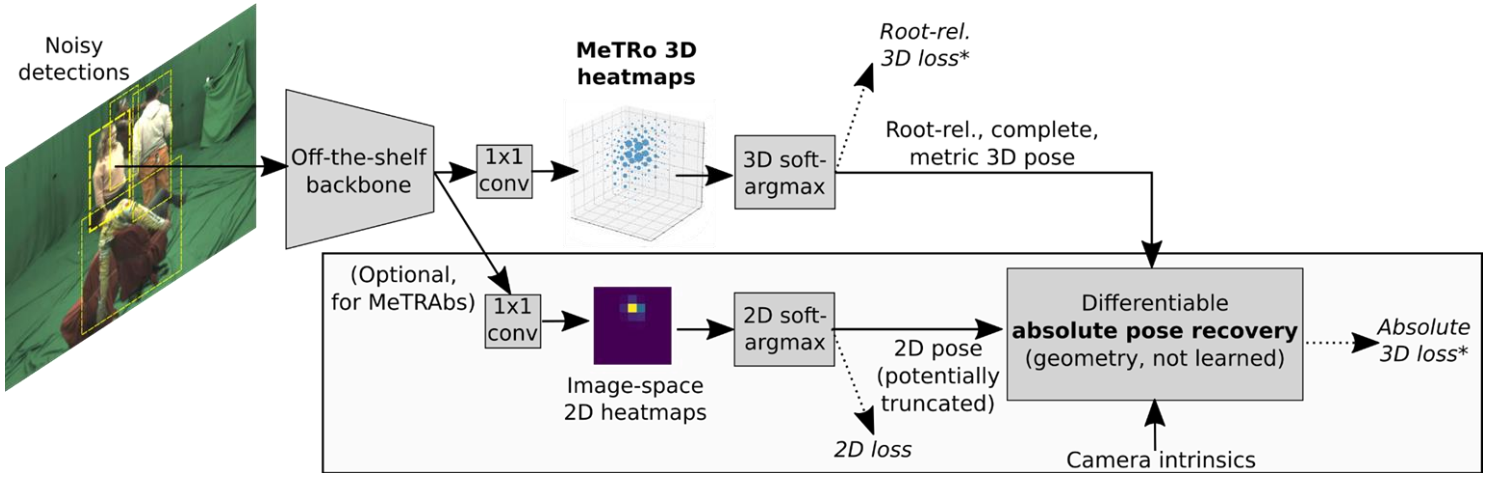
Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, Yaser Sheikh
Carnegie Mellon University

Z. Cao, G. Hidalgo Martinez, T. Simon, S.-E. Wei, OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields, IEEE Trans. PAMI, 2019. (code available here)

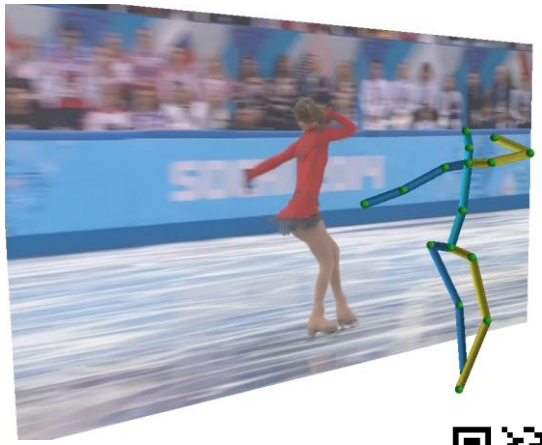
3D Human Body Pose Estimation

[Sarandi, Linder, Arras, Leibe, IEEE TBIOM'21]

Architecture: CNN + Differentiable Geometry



- 3D Body Pose Estimation
  - Metric-Scale Truncation-Robust Heatmaps for 3D human pose estimation (MeTRo)
  - Robust to truncation, suitable for close range
  - Very simple network design, extremely fast (can process >500 detections/s on a desktop GPU)



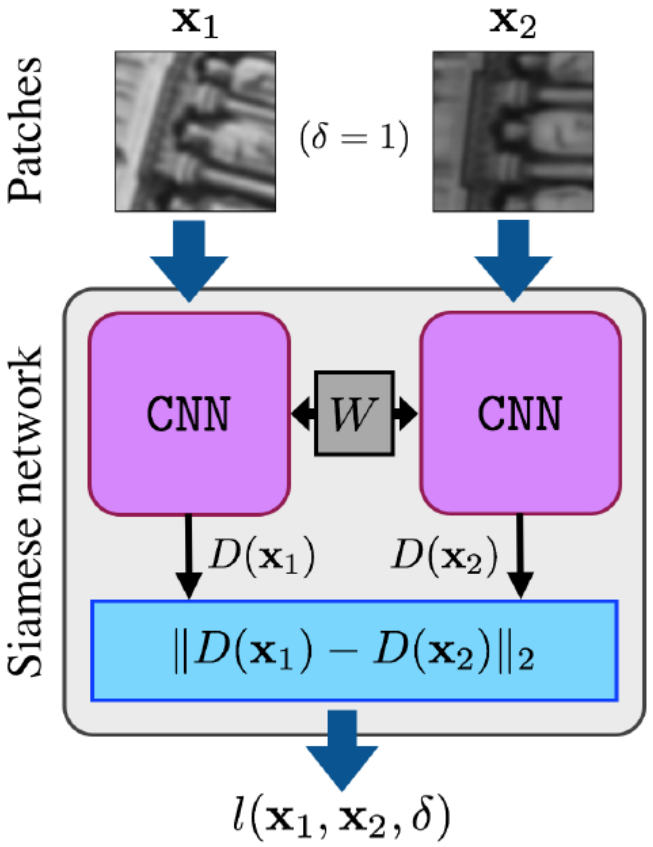
I. Sarandi, T. Linder, K.O. Arras, B. Leibe, MeTRAbs: Metric-Scale Truncation-Robust Heatmaps for Absolute 3D Human Pose Estimation, IEEE Trans. BIOM, 2020.

Topics of This Lecture

- CNNs for Object Detection
  - The R-CNN Family
  - The YOLO Family
- CNNs for Segmentation
  - Fully Convolutional Networks (FCN)
  - Encoder-Decoder architecture
  - Transpose convolutions
  - Skip connections
- CNNs for Human Pose Estimation
  - 2D Body pose estimation
  - 3D Body pose estimation
- CNNs for Matching
  - Siamese networks
  - Triplet loss

Learning Similarity Functions

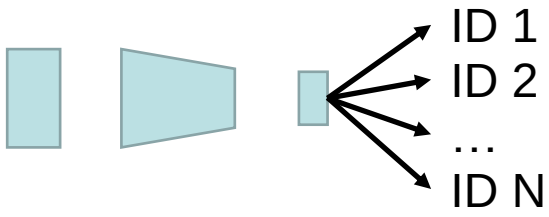
- Siamese Network
  - Present the two stimuli to two identical copies of a network (with shared parameters)
  - Train them to output similar values if the inputs are (semantically) similar.
- Used for many matching tasks
  - Face identification
  - Stereo estimation
  - Optical flow
  - …



- *How can we train a network to achieve this?*

Types of Models used for Matching Tasks

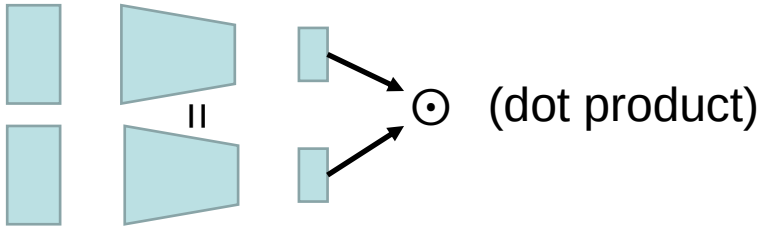
- Identification models (I)



Training

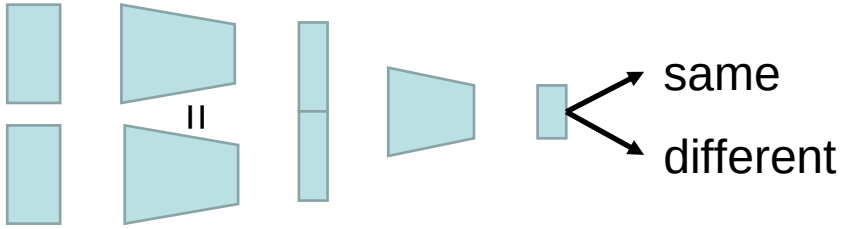
Multi-class classification loss

- Embedding models (E)



Large-margin loss, Triplet loss

- Verification models (V)



Two-class classification loss

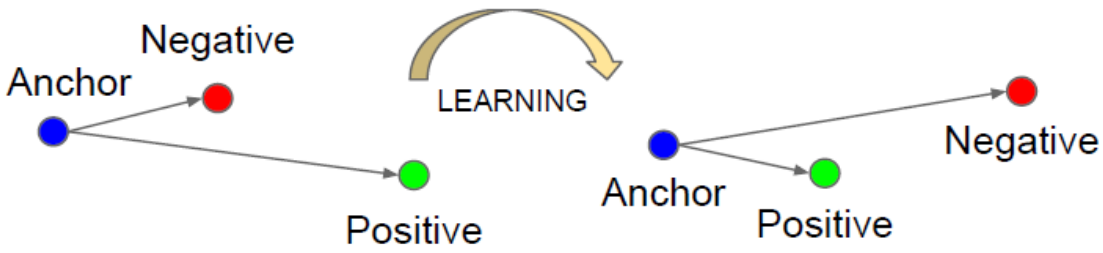
Triplet Loss Networks

- Learning a discriminative embedding
  - Present the network with triplets of examples

Negative Anchor Positive

  - Apply triplet loss to learn an embedding $f(\cdot)$ that groups the positive example closer to the anchor than the negative one.

$$\|f(x_i^a) - f(x_i^p)\|_2^2 < \|f(x_i^a) - f(x_i^n)\|_2^2$$



$\Rightarrow$ Used with great success in Google's FaceNet face identification

Triplet Loss – Practical Implementation

- Triplet loss formulation (using distances $D_{a,p} = d(x_a, x_p)$)

$$\mathcal{L}_{\text{tri}}(\theta) = \sum_{\substack{a,p,n \\ y_a = y_p \neq y_n}} [m + D_{a,p} - D_{a,n}]_+$$

- Practical Issue: How to select the triplets?
  - The number of possible triplets grows cubically with the dataset size.
  - Most triplets are uninformative

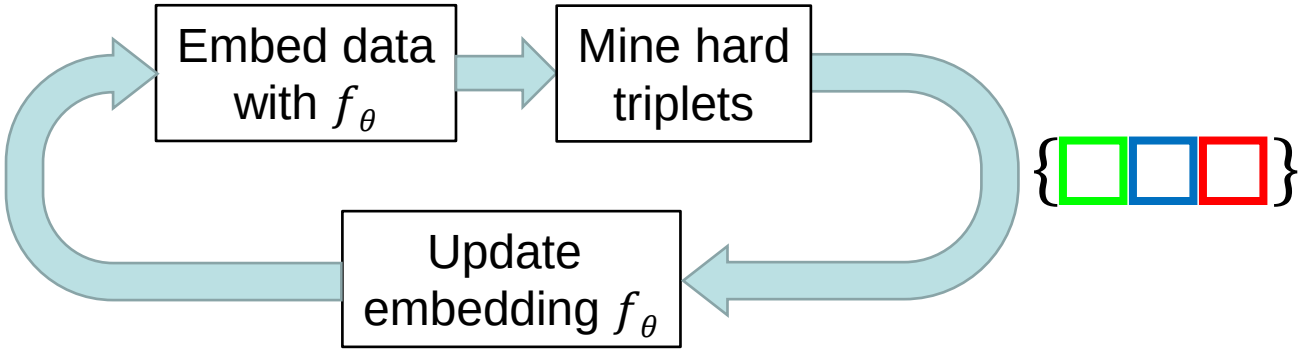
  $\Rightarrow$ Mining hard triplets becomes crucial for learning.

  $\Rightarrow$ Actually want *medium-hard* triplets for best training efficiency

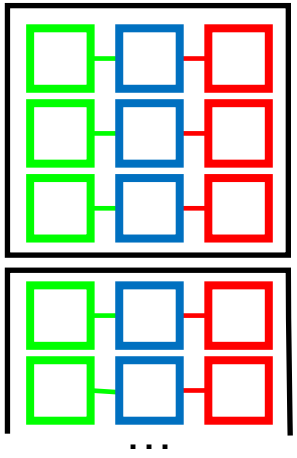
- Popular solution: Offline hard triplet mining
  - Process the dataset to find hard triplets
  - Use those for learning
  - Iterate

Offline Hard Triplet Mining

- Considerable effort needed



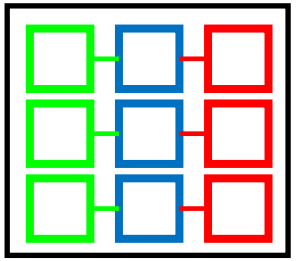
- Using the triplets for learning
  - Minibatch learning



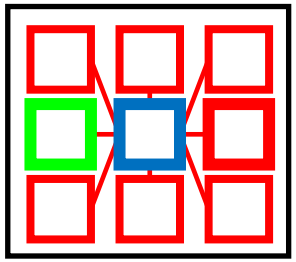
This is a very wasteful design!

Better: Online Hard Triplet Mining

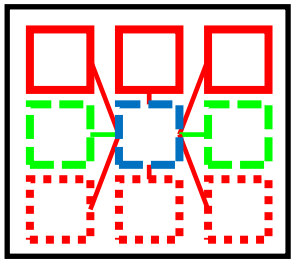
- Core idea
  - The minibatch contains many more potential triplets than the ones that were mined!
  - Why not make use of those also?



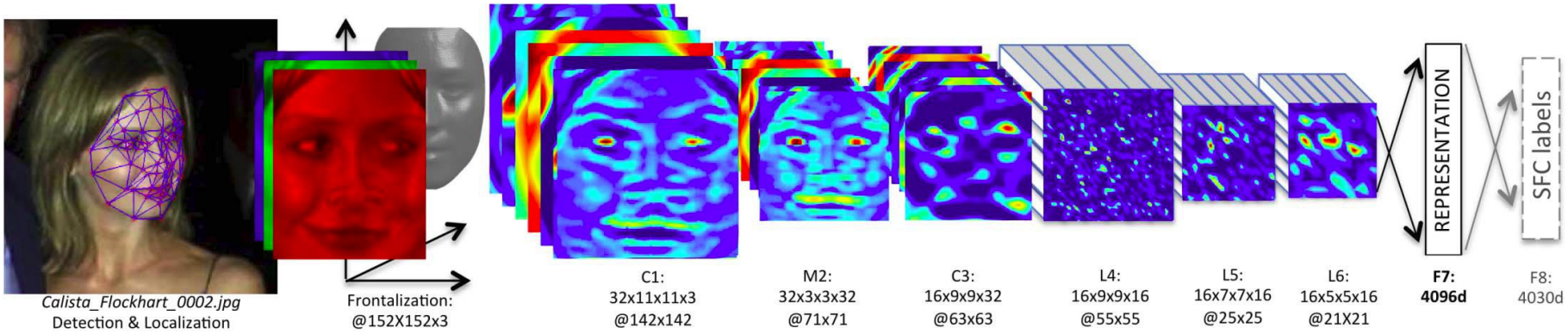
- Possible improvement
  - Each member of another triplet becomes an additional negative candidate
  - But: need both hard negatives *and* hard positives!

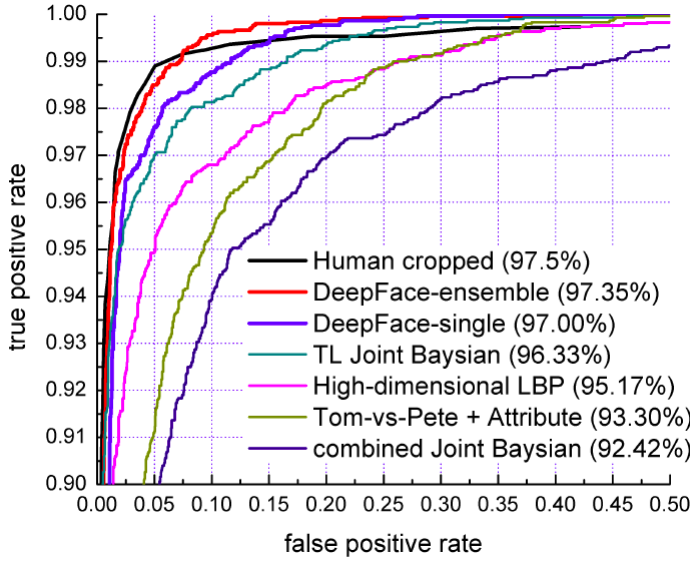


- Better design
  - Sample K images from P classes (=people) for each minibatch
  - Triplets are only constructed within the minibatch



Applications: Face Identification

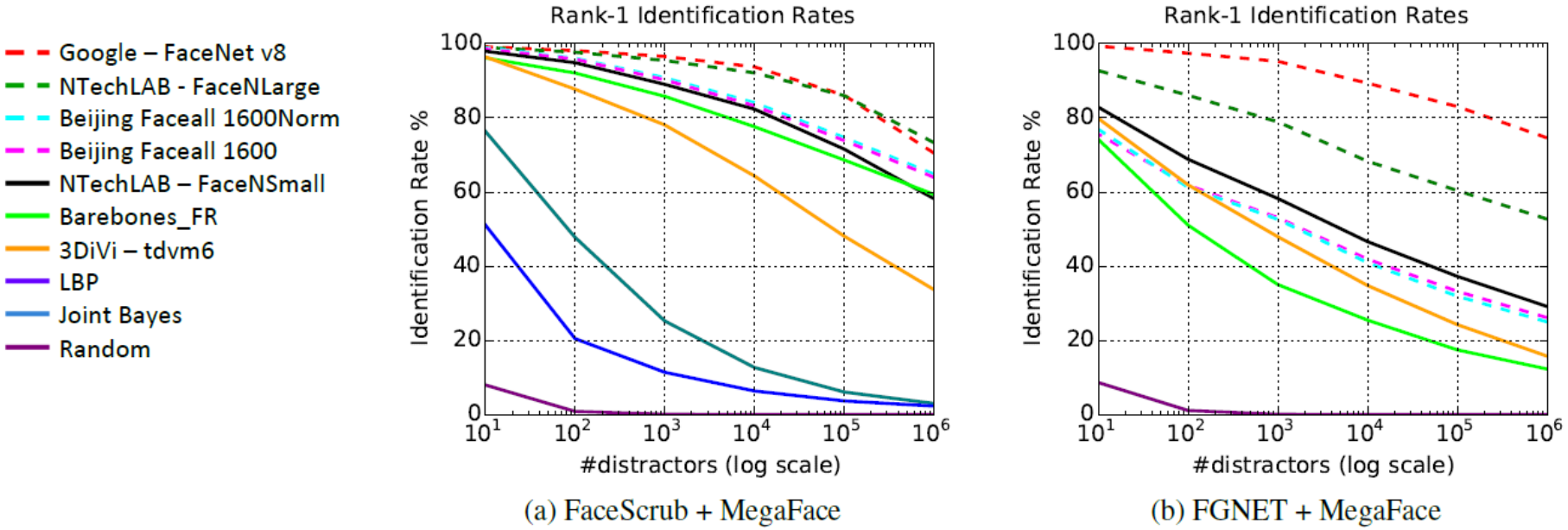




Y. Taigman, M. Yang, M. Ranzato, L. Wolf, DeepFace: Closing the Gap to Human-Level Performance in Face Verification, CVPR 2014

Slide credit: Svetlana Lazebnik

Face Identification Results

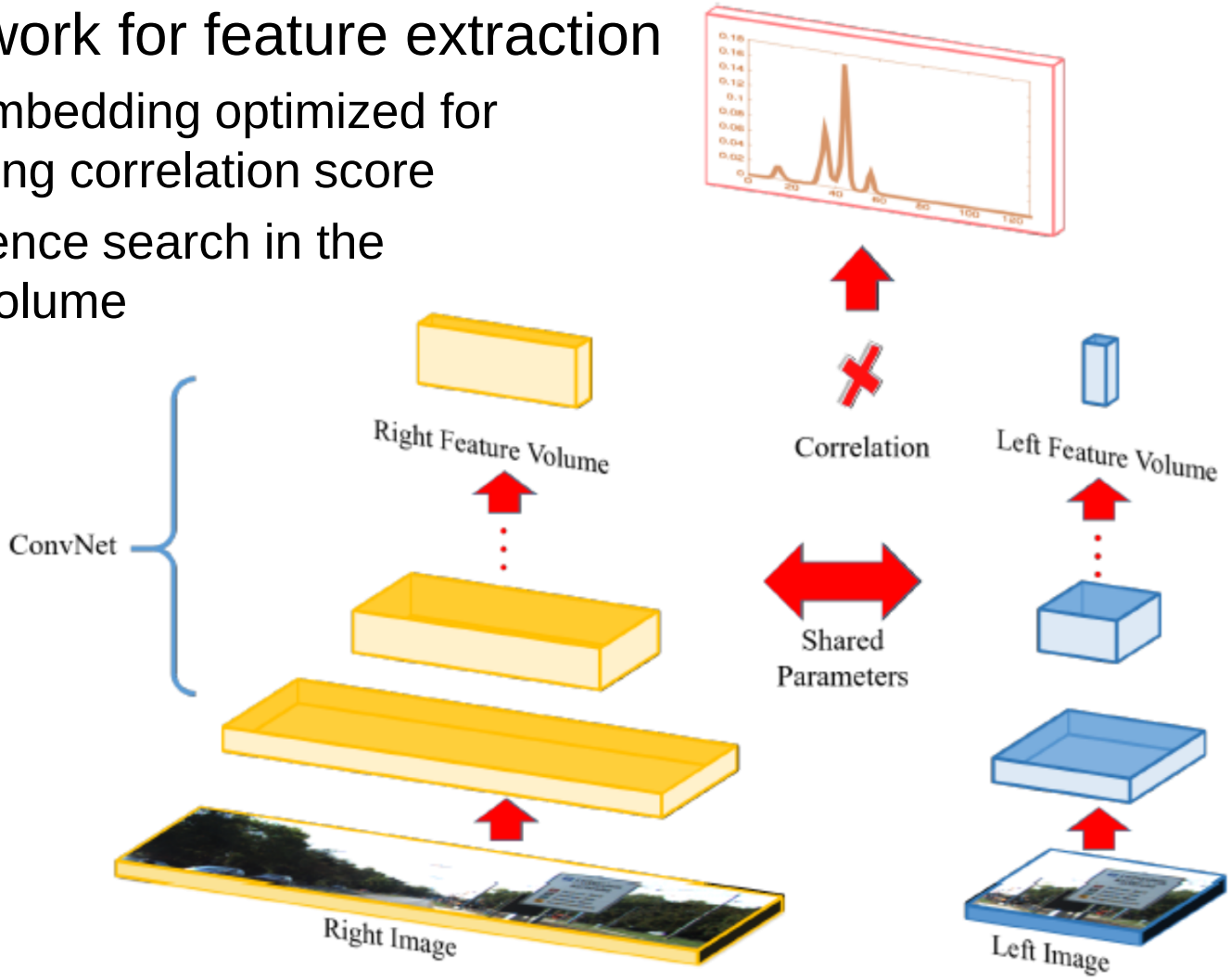


- Rank-1 identification rates of state-of-the-art algorithms
  - Performance with varying number of distractors in the gallery set
  - FaceNet: trained on 500M photos of 10M people

[Kemelmacher et al., CVPR'16]

Applications: Stereo Matching

- Siamese network for feature extraction
  - Learns an embedding optimized for matching using correlation score
  - Correspondence search in the correlation volume



[Luo, Schwing, Urtasun, CVPR'16]

References: Computer Vision Tasks

- Object Detection
  - R. Girshick, J. Donahue, T. Darrell, J. Malik, Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation, CVPR 2014.
  - S. Ren, K. He, R. Girshick, J. Sun, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NIPS 2015.
  - K. He, G. Gkioxari, P. Dollar, R. Girshick, Mask R-CNN, arXiv 1703.06870.
  - J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You Only Look Once: Unified Real-Time Object Detection, CVPR 2016.
  - W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C-Y. Fu, A.C. Berg, SSD: Single Shot Multi Box Detector, ECCV 2016.

References: Computer Vision Tasks

- Semantic Segmentation
  - J. Long, E. Shelhamer, T. Darrell, Fully Convolutional Networks for Semantic Segmentation, CVPR 2015.
  - O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional Networks for Biomedical Image Segmentation, MICCAI 2015
  - V. Badrinarayanan, A. Kendall, R. Cipolla, SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation, arXiv 1511.00561, IEEE Trans. PAMI 2017.
  - T-Y. Lin P. Dollar, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature Pyramid Networks for Object Detection, CVPR 2017.
  - L-C. Chen, G. Papandreou, F. Schroff, H. Adam, Rethinking Atrous Convolutions for Semantic Segmentation, arXiv 1706.05587 2017.

References: Computer Vision Tasks

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